TEMPLATE

KEY PROJECT INFORMATION & VPA DESIGN DOCUMENT (PDD)

PUBLICATION DATE **25/10/2022**

VERSION **v. 2.1**

RELATED SUPPORT Programme of Activity requirements and procedures

This document contains the following Sections

KEY PROJECT INFORMATION

Type of VPA	☐ Real case VPA (*An actual activity implemented/proposed to be implemented under a PoA that sets out a specific regulatory and design framework to be followed by similar regular VPAs. It is distinguished with other real case VPAs based on aspects like; technology/measure types, host country, end user type etc)* ☒ Regular VPA (*An activity involving single measure or a set of interrelated measures implemented under a PoA that follow the framework/requirements set out by an associated real case VPA and PoA)*
Scale of VPA Note that a VPA can be of one scale. Please select applicable scale accordingly.	☐Microscale ☒Small scale ☐Large scale *Refer to corresponding Activity Requirements for definition of the scale of PoA.* *A regular VPA shall comply with the scale requirements as defined at corresponding real case VPA.*
Title of corresponding real case VPA (if applicable)	NA
GS ID of real case VPA (if applicable)	NA
GS ID of VPA	GS 11245
Title of VPA	Community Carbon Safe Water Drinking Programme-VPA10
Time of First Submission Date	12/07/2021 and 9:00 pm
Date of Design Certification	26/04/2023
Version number of the VPA-DD	6
Completion date of version	18/08/2023
Coordinating/managing entity	UpEnergy Group
VPA Implementer (s)	Community Carbon

Project Participants and any communities involved	UpEnergy Malawi Limited
Host Country (ies)	Malawi
GS ID and Title of applicable Design Certified VPA	NA
GS ID and Title of applicable Performance Certified VPA	NA
Activity Requirements applied	☒ Community Services Activities *(in general – off grid renewable energy, distributed technology, biogas, WASH)* ☐ Renewable Energy Activities *(in general – renewable energy projects connected to national or regional grids or industrial facilities)* ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A *(projects that do not fall into either of the above, for example Shipping)* *Refer to the activity requirements for specific criteria*
Other Requirements applied	☒ Programme of Activity requirements and procedures *(PoA Requirements mandatory)* *Use this space to list requirements applicable to certain types of projects. E.g. microscale*
Methodology (ies) applied and version number	Emission Reduction from safe drinking water supply v1.0
Product Requirements applied	☒ GHG Emissions Reductions & Sequestration *(required to issue VERs/PERs and label CERs)* ☐ Renewable Energy Label *(required to label Renewable Energy Certificates)* ☐ N/A *(This is the rare case when a project chooses to issue neither emission reductions/labels or renewable energy labels)*
VPA Cycle:	☒ Regular *(Stakeholder Consultation (first round) has been conducted before the Start Date of the VPA)* ☐ Retroactive *(Stakeholder Consultation (first round) is conducted after the Start Date of the VPA)* *Retroactive VPAs can claim GS VERs/CERs for a max 2 years prior to the date of design certification.*

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in B.6)	Estimated Annual Average	Units or Products
13 Climate Action (mandatory)	Emission Reductions	26,477	tCO_2(eq)
1 (No poverty)	Percentage of users reporting money savings due to reduction in purchased fuel consumption in project	100	%
3 (Good Health and Well being)	Percentage of users reporting reduction in smoke/PM after shifting to ICS in project	100	%
5 (Gender Equality)	Percentage of users reporting time saving associated with cooking and fuel collection	95	%
6 (Clean water and sanitation)	Number of beneficiaries	108,300	-
8 (Decent Work and Economic Growth)	Total number of jobs created	40	-
12 (Responsible Consumption and Production)	Percentage of users reporting reduction in use of non-renewable biomass in the project scenario	100	%
15 (Life on Land)	Percentage of users reporting	100	%

Fuelwood
equivalent savings
in the project

SECTION A. DESCRIPTION OF PROJECT

A.1. Purpose and general description of project

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The purpose of the VPA is to provide in residential/institutional/commercial users with water purification system (WPS) such as Institutional water treatment (IWT) and Household water Treatment (HWT) technologies with an aim to provide safe drinking water and reduce/avoid greenhouse gas (GHG) emissions from the burning of non-renewable woody biomass and/or charcoal for cooking in Malawi.

The VPA is implemented by Community Carbon & UpEnergy Group is the Coordinating and Managing Entity (CME) of the PoA. Community Carbon will implement the programme in partnership with local partners and would ensure the last-mile distribution/installation of the water purification systems to the beneficiaries.

According to 2017 statistics, 844 million people still lacked even a basic drinking water service; and about 263 million people spent over 30 minutes per round trip to collect water from an improved source (constituting a limited drinking water service).[1]The lack of access to safe drinking water is very acute in sub-Saharan Africa with more than 92 million population still collecting drinking water directly from surface water sources.

Although 67 per cent of Malawi’s households have access to drinking water, distribution among districts, and between urban and rural areas, is uneven[2]. Improved drinking water sources are more common in urban areas at 87 per cent compared to 63 per cent in rural areas.[3] In rural areas, 37 per cent of households spend 30 minutes or more to

[1] https://www.who.int/news/item/12-07-2017-2-1-billion-people-lack-safe-drinking-water-at-home-more-than-twice-as-many-lack-safe-sanitation
[2] https://www.unicef.org/malawi/water-sanitation-and-hygiene
[3] https://www.unicef.org/malawi/water-sanitation-and-hygiene

fetch drinking water in comparison to 13 per cent in urban areas[4]. Further analyses within districts also reveals the distribution of water services in some areas is poor and uneven. Only 77 per cent of water points nationwide are functional. The rest no longer work because of old age, catchment deterioration, neglect, lack of spare parts and inadequate community-based water management structures. Women and children shoulder the burden of poor access to water services as they often walk long distances to collect water for their families[5]. Evidence shows that improving access to water significantly increases the time women spend time raising children and carrying out other household work, increasing their productivity and improving child wellbeing. Only 91% of the primary schools use a protected water source and 4.2% of schools have hand washing facilities.
The project aims at providing the users with water purification systems, thereby reducing the GHG emissions from the burning of non-renewable woody biomass and/or charcoal for treating the water.

A.1.1. Eligibility of the VPA under approved PoA

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No.	Eligibility Criterion	Description / Required condition	Description of the VPA in relation to the criteria, Means of Verification and Supporting evidence for inclusion
1	Location of the VPA	The VPA is located within the geographical boundary of one of the countries included in the PoA.	Institutional water treatment (IWT) and Household Water Treatment (HWT) technologies will be distributed to institutional/residential users within the geographic boundary of Malawi which is within the boundary of PoA.

[4] https://www.unicef.org/malawi/water-sanitation-and-hygiene
[5] https://www.unicef.org/malawi/water-sanitation-and-hygiene

2	Type of project	Projects will involve the distribution of energy-efficient cooking technologies or safe drinking water systems to residential/institutions/commercial users.	VPA includes distribution of HWT and IWT to institutional users. Detailed information on technology is given in Section A.3 of VPA-DD.
3	No Double counting of Clean Energy Products (CEPs) and VPAs within this PoA and across other PoAs	A unique numbering or identification system for the CEP installed is applied. This shall ensure no double counting of WPS within the PoA and ensure that WPS can be identified as belonging to this PoA and not to a PoA managed by any other CME.	Each WPS distributed under the PoA will have the UpE logo and unique serial number on the product. The unique serial number along with the customer details (name, address, is also stored in the sales database along with the unique serial number assigned to each product and the VPA assigned to it. CME has several mechanisms in place to ensure there is no double counting. At the point of sale, sales receipts are signed by the customers having a declaration that they are not part of any other programme. Customers are made aware about this clause before they sign the receipts. During the project implementation period, survey questionnaires have questions on end users being part of any other programme (ICS or WPS or Borehole projects). During the project implementation period, if it has been found that any of the end user is part of other project activity, that particular end user will not be considered in the ER calculation and will not be part

			of any of the project activity's ex-post surveys.
4	No Double counting of VPA	The VPA is exclusively bound to the PoA. The VPA shall not be proposed as an individual Gold Standard or CDM project and/or as a part of any other CDM PoA and/or any other mechanism to avail climate change mitigation benefits. A statement shall be included in the VPA-DD that the specific VPA will not be part of another single Gold Standard or CDM project activity or VPA under another PoA and confirmed by the Partner Organization (PO) implementing the VPA.	The carbon standard registries including UNFCCC, GS and VERRA have been checked and it is confirmed that the VPA has not been registered as an individual Gold Standard or CDM project and/or as a part of any other CDM PoA and/or any other mechanism to avail climate change mitigation benefits.
5	Awareness and agreement of those operating a VPA on PoA subscription	Contractual provisions to ensure that those operating the VPA are aware and have agreed that their activity is being subscribed to the PoA. In the case that the CME is not responsible for implementing the VPA, the organization responsible for VPA implementation has signed a contractual agreement with the CME to participate in the PoA. This agreement: Defines the ownership of the carbon emission reduction rights - Covers the distribution and monitoring related	Community Carbon will be implementing the project which is also the CME of PoA.

		responsibilities of the parties involved - Confirms that the CEPs to be distributed under the VPA have not and will not be distributed under any other carbon project (CDM project, PoA or Gold Standard project) - Cedes the rights to the carbon credits generated from VPAs under the PoA to the CME.	
6	Non-diversion of ODA in case of public funding	The CME and the VPA operator (in case of being different from the CME) shall confirm that there is no public funding or in the case of public funding, the annex I party will confirm that funding is not a diversion of Official Development Assistance.	It is confirmed that there is no diversion of ODA. An ODA declaration confirming the same has been submitted to GS.
7	VPA Crediting Period	VPA crediting period not to exceed the PoA end date and the starting date of the crediting period of a VPA shall be on or after: (i) The date of registration of the PoA, if the corresponding VPA-DD is submitted together with the request for registration; (ii) The date when the VPA was included in the PoA.	The crediting period of the VPA is 14/09/2022. The VPA will have a crediting period of 5 years which can be renewed twice, i.e. in total a maximum period of 15 years. The VPA crediting period will not exceed the end date of the registered PoA.
8	Approval of VPA by CME	CME approves each VPA to be included into its registered PoA.	Statement of CME in each VPA-DD giving approval for the VPA to be included into its registered PoA.
9	Methodology Requirement	Each VPA will comply with the applicability criteria of	The VPA complies with all applicability criteria of

		the METHODOLOGY FOR EMISSION REDUCTIONS FROM SAFE DRINKING WATER SUPPLY (Version 1.0)	METHODOLOGY FOR EMISSION REDUCTIONS FROM SAFE DRINKING WATER SUPPLY (Version 1.0). A detailed justification is provided in Section B.2 of VPA-DD.
10	Additionality	All VPAs to be included under the PoA will be in compliance with item 1.1.3 of Annex B – positive list mentioned in the 'Community Services Activity Requirements', Version 1.2. All VPAs will be solely composed of isolated units (CEPs) where the users of the technology/ measure are households or communities or institutions and where each unit results in <= a. 600 MWh of thermal energy savings per year. b. 600 tCO2 per year Hence, according to paragraph 4.1.9 of the 'Community Services Activity Requirements', each of the VPAs, regardless of the host country in which the project activity is being implemented, is deemed additional and therefore is not required to prove financial additionality at the time of Design Certification.	The emission reductions per year is below 60,000 tCO_2e per year. The same can be verified from ER Calculation sheet submitted with the VPA-DD.

11	Scale of the VPA	VPAs included under the PoA can be either small or large scale. In case of large scale VPAs, CME shall confirm that there is no suppressed demand claim. In case of small-scale threshold of 60,000 tCO_2/crediting year for safe water VPAs and 180 GWhth for ICS VPAs will be adhered.	This is safe water small-scale VPA emission reductions will capped at 60,000 tCO_2/crediting year.
12	Project Technology	Specification of technology or measures, such as the level and type of service, as well as performance specification based on, intra alia, testing/certification	Project technology along with technical specifications is outlined in section A.3
13	Sampling	Sampling approaches are set out in each VPA and will follow the TPDDTEC v3.1 and emission reduction from safe drinking water supply-version 1.0 methodology. For safe water cross-VPA sampling a maximum of 10 VPAs can be grouped.	Sampling method is described in VPA-DD as per applied methodology requirements. More than 10 VPAs will not be grouped in case of cross-VPA sampling.
14	Multi-country PoA	In accordance with 17.1.2 “Programme of Activity Requirements”, an exception is approved for the development multi-country PoA. The VPA-DD developed should be part of this approved deviation request (Deviation reference- “POA-17.003”) batching. As per 17.1.4 of “PoA Requirements”, one VPA shall be submitted at the time of design	The VPA-DD is developed for Malawi which is one of the countries in the approved deviation request “POA-17.003”.

		certification and subsequent VPAs for other countries shall be included at a later stage.	
15	SDG Outcomes	A minimum of two SDG outcome to be included in addition to SDG-13	All VPA-DDs will include a minimum of three SDG impact assessment including SDG-13. SDG targets and impact are described in section A.4 of the document

As per section 3.1.1 of GS4GG Principles & Requirements, compliance with relevant Eligibility criteria is demonstrated below:

Eligibility Criteria Category	Eligibility criterion - Required condition	Justification
1. Types of Project	Eligible projects shall include physical action/implementation on the ground. Pre-identified eligible project types are identified in the Eligibility Principles and Requirements section.	VPA is already implemented since 30/11/2020. Project is already one of the pre identified types as per section 3.1.1 (b) and automatically eligible for Gold Standard Certification as per section 4.1.3 of GS4GG Principles & Requirements.
2. Location of Project	Projects may be located in any part of the world.	Location of the VPA is the Republic of Malawi
3. Project Area, Project Boundary and Scale	The Project Area and Project Boundary shall be defined. Projects may be developed at any scale although certain rules, requirements and limitations may apply under specific Activity Requirements, Impact Quantification Methodologies and Products Requirements. In order to avoid double counting the Project shall not be included in any other voluntary or compliance standards programme unless approved by Gold Standard (for example through dual certification). Also, if the Project Area overlaps with that of another Gold Standard or other voluntary or compliance standard programme of a similar nature, the Project shall demonstrate that there is no double counting of	The boundary for the VPA in terms of a geographical area is defined as the territorial boundary of the Republic of Malawi. The VPA is implemented within the geographical boundary of the PoA. To avoid inclusion of any WPS which is a part of another registered carbon project/ programme, all WPS under this programme shall have a unique ID number / Tag number, either inscribed on the WPS or retained by the buyer, to uniquely identify the WPS avoiding any double counting and trace its user, later during monitoring and verification.

Eligibility Criteria Category	Eligibility criterion - Required condition	Justification
	impacts at design and performance certification (for example use of similar technology or practices through which the potential arises for double counting or misestimation of impacts amongst projects)	
4. Host Country Requirements	Projects shall be in compliance with applicable Host Country's legal, environmental, ecological and social regulations.	The VPA complies with Malawi's legal, environmental and ecological and social regulations. The VPA is in line with Malawi's national policies revolving around water access, rural water infrastructure, community engagement, women empowerment and climate change action. National Water Policy 2007 and Malawi Vision 2020.
5. Contact Details	As part of the Project Documentation the Project Developer shall provide (i) name and (ii) contact details of all Project Participants; AND in case of an organisation (iii) the legal registration details and (iv) documentation by the governing jurisdiction that proves that the entity is in good standing (defined as being a legal or other appropriate entity registered in or allowed to operate within the required jurisdiction and with no evidence of insolvency or legal/criminal notices placed against it or any of its Directors). Gold Standard retains the right (at its own discretion) to refuse use of the Standard where reputational concerns are highlighted.	Name and Contact details of Project Participants is given in the Appendix 1. Legal registration details: Community Carbon is duly incorporated under the provisions of the Companies Act 2001 as Category 1 Global Business Company by Republic of Mauritius on 03/06/2022. Company Registration No: 188181. Relevant document is submitted.
6. Legal Ownership	Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated. Where such ownership is transferred from project beneficiaries this must be demonstrated transparently and with full, prior and informed consent (FPIC). Note that for	Criteria for transfer of carbon credit ownership: • This is a retroactive VPA, shall ensure through relevant provisions for example disclaimer on warranty cards, stove packaging, customer agreements / sales receipts/ consent form or may be collected via

Eligibility Criteria Category	Eligibility criterion - Required condition	Justification
	certain Project types there is a requirement for full and uncontested legal land title/tenure to be demonstrated. These are contained within specific Activity or Product Requirements. All projects shall immediately report to Gold Standard any land title/tenure disputes arising.	monitoring app, etc. or stakeholder feedback collected during Stakeholder Feedback Round (SFR).
7. Other Rights	As well as legal title and ownership, the Project Developer shall also demonstrate where required uncontested legal rights and/or permissions concerning changes in use of other resources required to service the Project (for example, access rights, water rights etc.). Any known disputes or contested rights must be declared immediately to Gold Standard by the Project Developer and resolved prior to further project implementation in affected areas.	Not applicable
8. Official Development Assistance (ODA) Declaration	All Project Developers applying for project activities located in a country named by the OECD Development Assistance Committee's ODA recipient list and seeking Gold Standard Certification for carbon credits shall declare the Official Development Assistance (ODA) support. The Project Developer shall follow the GHG Emissions Reduction & Sequestration Product Requirements and submit the declaration at the time of Design Certification.	No ODA is involved in the VPA. A CME declaration is submitted.

Eligibility under Gold Standard Community Services Activity (CSA) Requirements

As per section 3 of GS4GG Community Services Activity (CSA) Requirements, Eligibility criteria is defined below:

Eligibility Criteria Category	Eligibility criterion - Required condition	Justification
1. **Eligible Project Types**	All CSA Projects shall lead to climate change mitigation and/or adaptation by providing or improving access to services/resources at the household or community or institution level. Eligible services include electricity and energy, water and sanitation, waste management, housing, etc.	The goal of the proposed VPA is to distribute WPS (improving access to services) in residential/institutional/commercial users within the national borders of the Malawi
2. **GENERAL ELIGIBILITY CRITERIA - Type of project**	(b) End-use energy efficiency: Project activities that reduce energy requirements as compared to baseline scenario without affecting the level and quality of services or products, where the end-user of the products and services are clearly identified and when the physical intervention is required at the user end. For example, efficient cooking, heating, lighting, etc.	The VPA involves distribution of safe water systems (HWT and IWT) to residential/ institutional users in Malawi.
3. **GENERAL ELIGIBILITY CRITERIA – Project Area, Boundary and scale**	Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements.	The project area is point location of WPS beneficiaries in the host country of the VPA. The project boundary will be limited to the geographical boundary of the host country
4. **GENERAL ELIGIBILITY CRITERIA – Legal Ownership**	(a) Projects involving the distribution of a large number of devices for services such as heating, cooking, lighting, electricity generation, water treatment technology such as water filter, etc. shall provide a clear description of the ownership of the Products that are generated under Gold Standard Certification all along the investment chain. In line with the FPIC requirement, the proofs that end-users are aware of and willing to give up their rights on Products shall be provided.	The CEP owners will be transferring their rights on ownership of carbon credits to CME via the end user agreement /consent form via monitoring app etc The same will be discussed during stakeholder consultations.

Eligibility Criteria Category	Eligibility criterion - Required condition	Justification
	(b) The transfer of Product ownership shall be discussed during local stakeholder consultations for projects.	

A.1.2. Legal ownership of products generated by the VPA and legal rights to alter use of resources required to service the project

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Community Carbon has the legal ownership of the Verified Emission Reductions (VERs) that are generated through the Gold Standard Certification. The carbon title for the product is signed off by end user directly to Community Carbon waiving any claim or rights on carbon credits generated under the VPA.

A.2. Location of VPA

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Host Party: Malawi

Region/State/Province: All across Malawi

City/Town/Community: All across Malawi

The geographical location of Malawi is depicted by the map below. Malawi, a landlocked country in southeastern Africa, is defined by its topography of highlands split by the

Great Rift Valley and enormous Lake Malawi with latitude 13.2534° S and longitude 34.3015° E.

Figure :-

The physical/geographical boundary of the SSC-PoA: Malawi

A.3. Technologies and/or measures

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The VPA will include following distribution of Safe drinking water system (WPS) to both residential and institutional Users. The programme will replace no boiling or conventional treatment systems of boiling using three stone fire or conventional system for woody biomass lacking improved combustion air supply mechanism and flue gas ventilation system, other conventional system using woody biomass, improved cookstoves with thermal efficiency ≥ 20%, and fossil fuel combusting systems in the baseline. The water purification treatment options will be selected to ensure that they provide safe and hygienic potable water. The chosen technologies will be in compliance with the host country norms.

Sustainable Development Goals Targeted	How the project contributes to the identified SDG
13 Climate Action (mandatory)	There are no emissions from WPS therefore reducing the GHG emitted.
1 No Poverty	Projects leads to money saving from less fuel/no fuel consumption in the cleaner technology
3 Good Health and Wellbeing	The project results in providing access to safe drinking water therefore improving the health of the end-users
5 Gender Equality	Project alleviates the strain on women by reducing the time spent on boiling water and fuel collection
6 Clean Water and Sanitation	Project provides access to safe drinking water technology
8 Decent Work and Economic Growth	The project involves distribution of water purification technology which in turn means number people will be employed to make the programme a success.
12 Responsible Consumption and Production	Project results in reduction in fuel consumption as compared to baseline
15 Life on Land	Project results in reduction in fuel consumption as compared to baseline which in turns reduce deforestation

Examples of a Water filters that shall be included in this programme is provided below[6]:

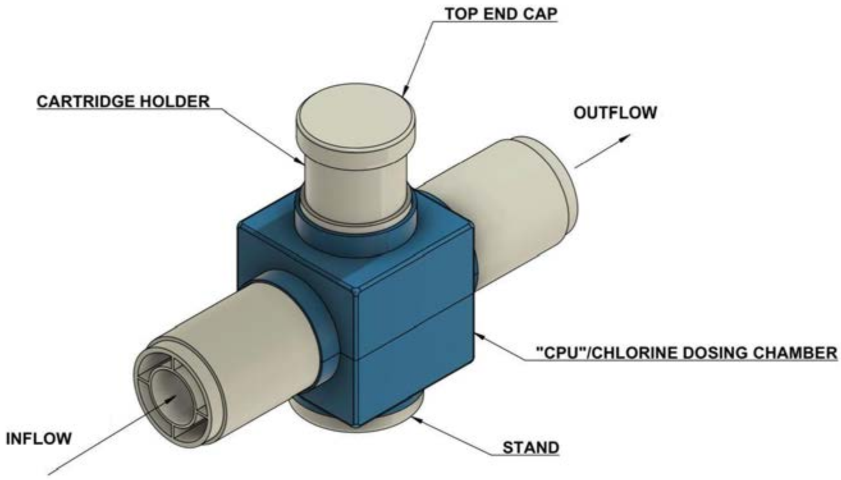
1. Community Carbon JH-19B Gravity Water filter
    - Dimension: (31 x 31x62)cm
    - Weight: 3.05 Kilograms
    - Capacity: 19 Liters
    - Flow Rate: 1 Liter per hour.
    - Materia: ce, EMC, RoHS
    - Filter life/Capacity: 2400 Liters

2. Community Carbon PEHF521 Gravity Water Filter
    - Dimension: (25 x 25 x 60) cms
    - Weight: 2.1 kilograms
    - Flow Rate: 2 Liters/Hour
    - Material: Polypropylene plastic
    - Filter Life/ Capacity:5000 Liters (Best case)

[6] This is an indicative list of products, other models may be added during the implementation of VPA.

- Certification: CNAL, US EPA, SGS

3. PurAll 50
   - Dimension: (30 x 10 x 25) cm
   - Flow Rate: 50 Liters/Minute
   - Material: Plastic
   - Filter Life/ Capacity:125000 Liters (Best case)
   - Certification: Certified to NSF ANSI Standard 60



The replacement filtration parts will be provided to cater based on the consumption need beyond the life of the product.

A.4. Scale of the VPA

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The VPA is a small-scale project activity. The project at any crediting period will not exceed threshold of 60,000 tCO_2 in any crediting year. This will be demonstrated in ER calculation sheet.

A.5. Funding sources of VPA

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There is no public funding for the PoA. No ODA funding will be used, as confirmed by signed ODA Declarations submitted to GS.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

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Emission Reduction from safe drinking water supply version 1.0

Requirements and Guidelines for carrying out usage surveys for projects implementing improved cookstoves version 2.

Tool:-

CDM Methodological tool 30: Calculation of the fraction of non-renewable biomass, Version 03.0.

B.2. Applicability of methodology (ies)

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The table below lists each of the methodology applicability requirements and how the VPA complies with each of the criteria:

S.No	Methodology applicability requirement	Justification
1	This methodology is applicable to the PoA introducing household water treatment technologies (HWT), institutional water treatment technologies (IWT), and community level water treatment technologies (CWT) include bleach/chlorine,	This VPA includes distribution of HWT and IWT to reduce or avoid GHG emission from boiling unsafe drinking water in the baseline and lacking access to safe drinking water (suppressed demand) to households and Institutions who are the

	water filter (ceramic, sand, composite, membrane, etc.), UV disinfection, etc.	target end users. In the absence of this VPA, the above two are the baseline methods for drinking water in households as well as institutions.
2.	Eligible community water supply technologies (CWS) include new installation of new borehole hand-pumps, borehole hand-pumps rehabilitation, solar powered drinking water pumps, etc. Water pumps powered by fossil-fuel engines are not eligible, with the exception of backup fossil–fuel engines that are used for no more than 10% of operating hours	Not Applicable in this VPA as Community Carbon is not including community water supply technology (CWS).
3.	.Eligible CWT and CWS technologies include ongoing maintenance and repair of the project technology	Not Applicable in this VPA as Community Carbon is not including CWS and CWT.
4.	The project involves the rehabilitation of an existing technology, the project developer shall provide evidence that the existing technology is non-operational and that there is no planned maintenance or repair for at least 3 months after the date it became non-operational	Not Applicable as this is applicable to CWS and CWT technologies.
5.	The methodology allows for project activities to include safe water treatment and/or supply technologies implemented for end-users in households, and/or commercial premises such as shops or institutional premises including half or full day/boarding schools, prisons, army camps & refugee camps.	This VPA includes distribution of HWT and IWT to reduce or avoid GHG emission from boiling unsafe drinking water in the baseline and lacking access to safe drinking water (suppressed demand) to households and Institutions who are the target end users.
6.	Demonstration of safe water is retrieved at the CWT or CWS location, the water in its improved form shall be available within a distance of 1 km or less from the end-users by satellite imaging or GPS coordinates of each CWT or CWS location. Alternatively, to demonstrate, as a proxy, a total collection	Not Applicable as this is applicable to CWS and CWT technologies.

	time of 30 minutes or less for a round trip, including queuing, using the travel modes of walking or pedaling.	
7.	Demonstration of Project technology performance level of HWT and IWT: It shall be demonstrated based on report of laboratory testing or official notification that the project technology or equipment achieves either (i) the performance target classification 3-star or 2-star level, meaning “Comprehensive Protection,” as per the WHO International Scheme to Evaluate Household Water Treatment Technologies (World Health Organization, 2011) or (ii) compliance with the national standard or guideline for household drinking water treatment technology; if no national guideline or standard is available, then the project technology shall comply with the WHO International Scheme requirements as per (i)	As an example, Community Carbon will distribute few models like Community Carbon SmartHomeJH-19B Gravity water filter, SmartHome PEHF521 Gravity water filter, PurAll50with different capacities which will comply with WHO Quality Standards or National Standards of the host country. The third-party certification by a qualified entity, which is a recognized certification agency by National/ International Standard body will be demonstrated for various safe drinking parameters. The technical specifications of example models are given in above section A 3. These are the example models and additional models may be added more HWT and/or IWT technologies which comply with WHO or National standards.
8.	Demonstration of Project technology performance level of CWT and CWS: For each individual CWT or CWS, it shall be demonstrated at the start of each crediting period with water quality testing reports that the water directly supplied by the project water technology/source achieves both: a. microbial quality in line with either (i) national standards or guidelines for microbial quality of drinking water, or in the absence of such requirements, (ii) the guideline values for verification of microbial quality from the Guidelines for drinking-water quality	Not Applicable as this is applicable to CWS and CWT technologies which is not part of this VPA.

	b. compliance with (i) national standards or guidelines on priority chemical contamination and physical and aesthetic aspects, or in the absence of such requirements, (ii) international standards or guidelines on priority chemical contamination11 and physical and aesthetic aspects.	
9.	To conduct annual water hygiene education campaigns for the end-users in this project.	Annual water hygiene education campaigns will be conducted. During monitoring of households and Institution, CME shall conduct a representative sample survey annually and will be reported as “report of annual hygiene campaign results” and summarized in the monitoring report. Any major change will be reported and strategy will be addressed through subsequent health campaign. The comprehensive steps or methods to access hygiene handling of clean water will be provided in project design document.
10	A project applying this methodology may make SDG claims if relevant monitoring parameter(s) is included in the monitoring plan to demonstrate and confirm the project’s contributions to SDGs 12. See parameter SDWS 19.	The project developer /CME will capture all the SDG indicators which is relevant to this project through monitoring in both HH and Institutions. The monitoring will be done using a detailed questionnaire which includes all the SDG indicators. For example, capturing water quality.

B.3. VPA boundary

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The project boundary is the physical, geographical sites of the project technologies, which is the location of the users included in the VPA-DD.

The greenhouse gases included in or excluded from the project boundary are shown in below table

Source		GHGs	Included?	Justification / Explanation
Baseline scenario	Emissions from non-renewable biomass utilized for obtaining safe drinking water displaced due to project activity.	CO_2	Yes	Major source of emissions
		CH_4	Yes	Minor source of emission
		N_2O	Yes	Minor source of emission
Project scenario	Emissions from electricity for operating project water supply/treatment technology	CO_2	No	Limited electrical energy may be required
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification
	Emissions from fossil fuels for operating project water supply/treatment technology	CO_2	No	Limited fuel energy may be required
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification

B.4. Establishment and description of baseline scenario

>>

Malawi has seen an increase of droughts and floods in recent years. In Malawi, 86% of households have access to an improved source of drinking water[7] 99% of urban households and 84% of rural households have access to improved water sources. Urban and rural households rely on different sources of drinking water. 46% of urban

[7] https://dhsprogram.com/pubs/pdf/FR319/FR319.pdf?msclkid=1d89bba5bf1511ecab544783dd3a34cc

households have piped water in their dwelling or yard, which accounts for the largest percentage of improved water sources for urban households. In contrast, rural households with access to improved sources of drinking water rely mainly on tube wells or boreholes (65%). While 91% of urban and 66% of rural households have water on the premises or travel less than 30 minutes to fetch drinking water[8].

The most common sources of drinking water in urban households are water piped into the household's dwelling, yard, or plot (45.7%); water from a public tap or standpipe (34.2%); and water piped to a neighbour (8.7%)[9]. Rural households obtain their drinking water mainly from tube wells or boreholes (65.4%), followed by protected dug wells (6.2%). 16.3% of rural households obtain their drinking water from an unimproved water source, as compared with 1.4% of urban households[10].

Further analyses within districts also reveals the distribution of water services in some areas is poor and uneven. Only 77 per cent of water points nationwide are functional. The rest no longer work because of old age, catchment deterioration, neglect, lack of spare parts and inadequate community-based water management structures.Women and children shoulder the burden of poor access to water services as they often walk long distances to collect water for their families. Evidence shows that improving access to water significantly increases the time women spend time raising children and carrying out other household work, increasing their productivity and improving child wellbeing. Appropriate methods of water treatment include boiling, bleaching, filtering, and solar disinfecting.

As per DHS MMIS report 2017, The proportion of households obtaining water from improved sources increased slightly from 81% in the 2012 to 83% in the 2014, and then to 86% in the 2017. The increase occurred in both urban and rural households, from 93% in 2012 to 99% in 2017 in urban areas and from 79% in 2012 to 84% in 2017 in rural areas.

The comprehensive questionnaire was designed to capture all the information of household like family members, age group, gender, address, contact number,

[8] https://dhsprogram.com/pubs/pdf/FR319/FR319.pdf?msclkid=1d89bba5bf1511ecab544783dd3a34cc
[9] https://dhsprogram.com/pubs/pdf/FR319/FR319.pdf?msclkid=1d89bba5bf1511ecab544783dd3a34cc
[10] https://dhsprogram.com/pubs/pdf/FR319/FR319.pdf?msclkid=1d89bba5bf1511ecab544783dd3a34cc

geographical coordinates, water collection, water source, consumption, health relates issues, etc. The household surveyed were not improved source of drinking water.
Clean, improved and safely managed drinking water services have the opportunity to improve several parts of a population's health and well-being.
The baseline scenario as per the methodology for emission reduction from safe drinking water supply, version 1.0, is defined by the typical baseline safely managed drinking water patterns in a population that is targeted for the adoption of the project technology.
The baseline survey covered the following elements:

- Water consumption pattern.
- Available improved and safe technology options used for water filtration
- Classified the most promising water treatment method for consumption.
- Estimated the potential of safe water filters obtained by using improved technologies and identify the profitability of these technologies.

The survey in person interview was conducted in various households in Malawi. The survey is designed to provide estimates at the national level, for urban and rural areas, and for the water consumption pattern. According to the baseline survey the average household size in Malawi is 4.5 persons.
The target population for this VPA are all users in household and/or institutions of either traditional biomass stoves, three stone fired that have baseline stove efficiency of 10% (or less), other conventional system using woody biomass with efficiency 20%, improved cookstove with default efficiency 30%, and fossil fuel combusting system. Suppressed demand can be applied in instances where inadequate safe water is available or where treatment is not practiced. Therefore, only users that boil water or are currently using unsafe water are eligible for crediting.

As per study conducted by Water Aid, in Malawi, more than 3100 children under five die a year from diarrhea. Globally, illnesses due to contaminated water cause children to lose 443 million school days each year. Even though access to improved sources are improving, yet 19% of the schools drink from unprotected sources.

This VPA will introduce safe drinking water system, with higher efficiency of water filtration to residential/ institutional/ commercial users by leveraging resources provided by the PoA. Therefore, it is assumed that in the absence of the project activity, the unimproved drinking water usage and appropriate drinking water treatment method for meeting similar safe water services needs share.

B.5. Demonstration of additionality

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Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).	Community Services Activity Requirements (Version 1.2), paragraph 4.1.9: “Projects that meet any of the following criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of Design Certification: (a) Positive list (Annex B) (b) Projects located in LDC, SIDS, LLDC Micro-scale projects”
Describe how the proposed project meets the criteria for deemed additionality.	Malawi is an LDC/LLDC, thus deemed additional

B.5.1. Prior Consideration

>>

Not Applicable

B.5.2. Ongoing Financial Need

>>

Not applicable.

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact Indicator (Proposed or SDG Indicator)
13 Climate Action (mandatory)	N/A	Emission Reductions
1 End poverty in all its forms everywhere	1.4 By 2030, ensure that all men and women, in particular the poor and the vulnerable, have equal rights to economic resources, as well as access to basic services, ownership and control over land and other forms of property, inheritance, natural resources, appropriate new technology and financial services, including microfinance	Percentage of users reporting money saving due to reduction in purchased fuel consumption in the project
3 Good Health and Well-being	3.3 By 2030, end the epidemics of AIDS, tuberculosis, malaria and neglected tropical diseases and combat hepatitis, water-borne diseases and other communicable diseases	Percentage of users confirming access to safe water using clean technology
5: Gender Equality	5.4 Recognize and value unpaid care and domestic work through the provision of public services, infrastructure and social	Percentage of users reporting time saving associated with cooking and fuel collection

	protection policies and the promotion of shared responsibility within the household and the family as nationally appropriate	
6: Clean Water and Sanitation	6.1 By 2030, achieve universal and equitable access to safe and affordable drinking water for all	Number of beneficiaries
8: Decent Work and Economic Growth	8.3 Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises, including through access to financial services	Total no of jobs created
12: Responsible Consumption and Production	12.2 By 2030, achieve the sustainable management and efficient use of natural resources	Percentage of users confirmed less use of fuel for boiling
15: Life on Land	15.2 By 2020, promote the implementation of sustainable management of all types of forests, halt deforestation, restore degraded forests and	Percentage of users reported Fuelwood(eq.) savings in the project scenario

	substantially increase afforestation and reforestation globally

B.6.1. Explanation of methodological choices/approaches for estimating the SDG Impact

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The VPA includes distribution of HWT and IWT with the aim to introduce zero or low GHG water purification systems to provide safe drinking water to households and/or institutional users such as schools with the aim to provide safe drinking water systems and reduce greenhouse gas (GHG) emissions from the burning of non-renewable woody biomass and/or charcoal for water treatment in Malawi.

The outcome of the SDG 13 (Climate Action) will be measured as reduced greenhouse gas emissions measured as tonnes of CO_2e applying the GS methodology "Emission Reductions from Safe Drinking Water Supply v1.0". The SDG 13 outcome will be certified as "Certified SDG 13 Impacts" allowing the generation of carbon credits (GS VERs). The overall GHG reductions achieved by the project activity in year y are calculated as follows:

Baseline Scenario Fuel Consumption Calculation

The total safe water consumed in the project scenario is the amount of safe water supplied by the project technology and consumed in the project scenario. This total is assumed to be equivalent to water boiled in the baseline. If the total volume exceed the cap stipulated in the table located in the section on suppressed demand, the claim for emission reductions . In case the CME wishes to monitor the parameter, will ensure the water consumption (per capita) doesn't exceed the capped value of 5.5 litre/person/day[11].

[11] CAP value is determined based on https://www.who.int/water_sanitation_health/diseases/WSH03.02.pdf

The baseline emission factor shall be calculated as

$SEw,b,y = 360.83/\eta_{wb}$

Where:

360.83	Default amount of energy required to obtain 1 L of water after 5 minutes of boiling from a first principles approach kJ/l
η_{wb}	Efficiency of the stoves for baseline water boiling (%). Weighted average of baseline stove types

The baseline emission shall be calculated as

$BEy = EFb \times (1 - Cb - Xcleanboil,y) \times Qy \times Mq,y$

Where:

BE_y	Baseline emissions from the use of fuel to obtain safe water in the baseline (tCO2e)
EF_b	Emission factor for the use of fuel to obtain safe water in the baseline (tCO2e/L)
C_b	Proportion of project end-users who in the baseline were already using a safe water supply that did not require boiling (%)
$X_{cleanboil,y}$	Proportion of project end-users that boil safe water in the project year y (%)
Q_y	Quantity of safe drinking water provided by the project in year y (L)
$M_{q,y}$	Modifier for the water quality in year y

The quantity of safe drinking water provided by the project is calculated using following method (for HWT and IWT)

The quantity of safe drinking water provided by the project Qy is determined as follows:

$Qy = \sum Np,y \times Up,y \times QPWhh,p,y \times DPp,y\ p$

Where:

Q_y	Quantity of safe drinking water provided by the project in year y (L)
$N_{p,y}$	Number of premises type p with at least one project technology in year y
$U_{p,y}$	Usage rate of the project technology by premises type p during year y (%)
$QPW_{hh,p,y}$	Volume of drinking water per premises p per day in year y (L)

$DP_{p,y}$	Days the project technology is present for end-users in the premises p in year y

The volume of drinking water per premises per day is determined by considering whether the capacity of the project device is sufficient to provide at least the default amount of drinking water, as follows:

$QPWhh,p,y = \min((qi \times tp,y \times DNp,y), (QPWp \times HNp,y))$

Where:

Project Scenario Fuel Consumption Calculation

Project emissions in year y (t CO2e/yr)

$PE_y = PE_{ff,p,y} + PE_{ec,p,y}$

Where:

PE_y	Project emissions in year y (tCO2)
$PE_{ff,p,y}$	Project emissions from fossil fuel use in year y (tCO2)
$PE_{ec,p,y}$	Project emissions from electricity use in year y (tCO2)

As the filters don't use fossil-fuel or electricity for filtration,

$PE_y = 0$

In case the CME introduces safe water technology which uses electricity in future, project emissions shall be calculated according to the methodology.

Leakage

Leakage ($LE_{p,y}$): As per applied GS Methodology for emission reductions from safe drinking water supply version 1.0, leakage emissions are accounted for the following sources:

#	Leakage Source	Assessment
a	Members of the population who do not participate in the project, and previously used lower emitting energy sources, instead use the nonrenewable biomass saved under the project activity.	No, as it is evident from the baseline survey that the share of household that use a lower emitting energy source such as LPG is very small. These households that use a lower emitting energy source, such as LPG, are not likely to use nonrenewable biomass

		(NRB) saved under the project activity, instead of LPG.
b	The project significantly reduces the NRB fraction within an area where other GHG mitigation project activities account for NRB fraction in their baseline scenario.	No, the project leads to reduction in firewood consumption which ultimately brings positive impact to the project vicinity area. It is not expected that the NRB in other areas will be affected so quickly that it would impact other CDM/VER projects activities if any. Furthermore, fNRB update takes place in each renewal of CP which automatically includes the studies evaluating the impact on fNRB.
c	The project population compensates for loss of the space heating effect of water boiling by adopting some other form of space heating or by retaining some baseline wood fuel-burning practices.	Not applicable, as the project population location is entire country of Uganda. Uganda is an African country crossed by the Equator, where the climate is pleasantly warm, with average temperatures ranging between 20 °C and 25 °C.

At the time of monitoring, PP shall be determining if the displaced technologies are used outside the project boundary in place of lower emitting technology through survey. As mentioned in the methodology, leakage is either calculated as a quantitative emissions volume (tCO_2e) or as a percentage of total emission reductions.

For ex-ante emission reduction estimation, leakage due to non-renewable biomass can be excluded as all the household supplied with safe water systems does not use lower-emitting energy sources and project activity doesn't increase NRB fraction.

LE_y = 0 (for ex-ante ER calculation)

Emission Reductions

The Emission reductions are calculated as follows:

$$ERy = BEy - PEy - LEy$$

Where:

ER_y	Emission reductions in year y (t CO2e/yr)
PE_y	Project emissions in year y (tCO2)
BE_y	Baseline emissions in year y (t CO2e/yr)
LE_y	Leakage emissions in year y (t CO2e/yr)

The other SDGs impacts of this VPA (SDG 1, SDG 3, SDG 5, SDG 6, SDG 8 SDG 12 and SDG 15) will be estimated qualitatively.

Approach to estimate SDG 1

SDG 1-NoPoverty		
$HHS_{baseline}$	% users reporting reduced money spent on fuel consumption in baseline scenario	0%
$HHS_{project}$	% users reporting reduced money spent on fuel consumption in project	100%
Percentage of users reporting money savings due to reduction in purchased fuel consumption in project	$(HHS_{project}\text{-}HHS_{baseline})$	100%

Approach to estimate SDG 3

SDG 3-Good health and well being		
$HHClean_{baseline}$	% users using clean technology to access safe water in baseline scenario	0
$HHClean_{project}$	% users using clean technology to access safe water in project scenario	100%
Percentage of users confirming access to safe water using clean technology	$(HHclean_{project} - HHclean_{baseline})$	100%

Approach to estimate SDG 5

SDG 5-Gender Equality		
$HHtime_{baseline}$	% users reporting time savings reported for boiling/fuel collection/access to water	0%
$HHtime_{project}$	% users reporting time saving reported due to reduced boiling/fuel collection/access to water	95%
Percentage of users reporting time saving associated with boiling and fuel collection/access to safe water	$(HHtime_{project}\text{-}HHtime_{baseline})$	95%

Approach to estimate SDG 6

SDG 6-Clean Water & Sanitation		
$BENclean_{baseline}$	Beneficiaries with access to safe water	0
$BENclean_{project}$	Beneficiaries with access to safe water due to project	1,083,00
Number of people/beneficiaries reporting access to safe water	$(BENclean_{project} - BENclean_{baseline})$	108300

Approach to estimate SDG 8

SDG 8-Decent work and economic growth		
$EG_{baseline}$	Generation of employment in baseline scenario	0
$EG_{project}$	Generation of employment in project scenario	40
Total number of jobs created	$(EG_{project} - EG_{baseline})$	40

Approach to estimate SDG 12

SDG 12 Responsible consumption and Production		
$FC_{baseline}$	%users confirming boiling using fuel in baseline scenario	100%
$FC_{project}$	% users confirming boiling using fuel in project scenario	0%
Percentage of users confirmed less use of fuel for boiling	$(FC_{baseline} - FC_{project})$	100%

Approach to estimate SDG 15

SDG 15 Life on land		
$FRC_{baseline}$	% users reported Fuelwood equivalent savings in the baseline scenario	0%
$FRC_{project}$	% users reported Fuelwood equivalent savings in the project scenario	100%
Percentage of users reported Fuelwood(eq.) savings in the project scenario	$(FRC_{project} - FRC_{baseline})$	100%

B.6.2. Data and parameters fixed ex ante

SDG13

Parameter ID	SDWS 2
Data/parameter	Project Technology Description
Unit	N/A
Description	The detailed description of the planned project technology shall include as a minimum: HWT and IWT: - manufacturer name, - product name (if applicable), - technology type, and - performance meets the WHO Quality standards or applicable national standards
Source of data	- - Commercial guarantee
Value(s) applied	HWT and IWT: - manufacturer name – Brand Name - product name - JH-19B Gravity Water filter, PEHF521 Gravity Water Filter and PurAll50[12] , - technology type – Gravity filters and Inline Chlorination System[13] - performance meets the WHO Quality standards
Choice of data or Measurement methods and procedures	-
Purpose of data	-
Additional comment	-

Parameter ID	SDWS 4
Data/parameter	Regulatory Framework for safe water supply

[12] Additional models may be offered during VPA implementation
[13] Other technology of safe water systems may be offered during VPA implementation

Unit	N/A
Description	List and provide a summary of any national, sub-national and local regulations or guidance for safe drinking water supply, operation and maintenance, including any tariff requirements.
Source of data	National, sub-national and local authorities
Value(s) applied	The project doesn't conflict with host country law. Please refer to Section B.4
Choice of data or Measurement methods and procedures	-
Purpose of data	-
Additional comment	-

Parameter ID	SDWS 5
Data/parameter	Water sources in the project boundary
Unit	N/A
Description	Identify the water sources in the project boundary, and identify whether they are used for drinking water, and for all that are used for drinking water, classify them as improved and unimproved water source.
Source of data	Any of the following sources shall be used: - Baseline study - Credible published literature for project region - Studies by academia, NGOs or multilateral institutions, or - Official government publications or statistics
Value(s) applied	<table><tr><th>Sources</th><th>Percentage</th></tr><tr><td>Unprotected Well</td><td>~14%</td></tr><tr><td>Municipality Water</td><td>~71%</td></tr><tr><td>Borewell</td><td>~13%</td></tr><tr><td>River</td><td>~1%</td></tr><tr><td>Pond/Lake/any stagnant body of water</td><td>~1%</td></tr></table>

Choice of data or Measurement methods and procedures	-
Purpose of data	-
Additional comment	-

Parameter ID	SDWS 6
Data/parameter	Stove technologies used in the project boundary
Unit	N/A
Description	The proportion of different stove types used in premises in the geographical area of the project.
Source of data	Any of the following sources shall be used: - Baseline survey, - Credible published literature for project region, - Studies by academia, NGOs or multilateral institutions, or - Official government publications or statistics
Value(s) applied	1. Three-stone fired firewood –27% 2. Traditional charcoal system without grate or/and chimney – 65%% 3. Improved cookstove – 8%% These are the prevalent technologies used by respondents who boil water in the baseline scenario
Choice of data or Measurement methods and procedures	-
Purpose of data	-
Additional comment	The classification shall consider at least the following categories of stoves types: - Three-stone fire or a conventional system for woody biomass lacking improved combustion air supply mechanism and flue gas ventilation system; - other conventional systems using woody biomass; - improved cookstoves (≥ 20% thermal efficiency); and

	- fossil fuel combusting systems

Parameter ID	SDWS 7
Data/parameter	Expected technical life of project technology
Unit	Volume or Years
Description	The expected technical life of an individual project technology shall be defined in the PDD. The details include both technology/device life and filter life, if a filter is used and it is replaceable.
Source of data	HWT/IWT: Any one of the following sources shall be used: - Manufacturer specifications - Third-party certification by a qualified entity, for example recognized certification agency by National/ International Standard body - Commercial guarantee
Value(s) applied	Please refer to Section A.3
Choice of data or Measurement methods and procedures	- JH-19B Gravity Water filter- 2400 Litres - PEHF521 Gravity Water Filter – 5000 Litres - PurAll50 – 125,000 Liters[14]
Purpose of data	-
Additional comment	

Parameter ID	SDWS 09
Data/parameter	EF$b,f,CO2$
Unit	tCO_2/TJ

[14] Additional models may be offered during VPA implementation

Description	CO_2 emission factor arising from use of fuels in baseline Scenario
Source of data	IPCC defaults for wood and charcoal, the following defaults derived from the IPCC shall be applied:
Value(s) applied	Wood: 112 tCO2/TJ Charcoal: 165.22 tCO2/TJ (includes charcoal production emissions)
Choice of data or Measurement methods and procedures	Default IPCC value for fuelwood and charcoal is applied
Purpose of data	CO_2 Emission calculation in baseline
Additional comment	-

Parameter ID	SDWS 10
Data/parameter	$EF_{b,f,non\text{-}CO2}$
Unit	tCO_{2e}/TJ
Description	Non-CO2 emission factor from use of fuels, in case the baseline fuel is biomass or charcoal
Source of data	IPCC defaults for wood and charcoal, the following defaults derived from the IPCC (AR 6)
Value(s) applied	Wood: 9.46 tCO2e/TJ Charcoal: 44.83 tCO2e/TJ (includes production emissions of CH4 and N2O)
Choice of data or Measurement methods and procedures	Default IPCC value for fuelwood and charcoal is applied
Purpose of data	Non-CO_2 Emission calculation in baseline
Additional comment	-

Data/parameter	ηwb
Unit	Percentage
Description	Weighted average efficiency of the baseline water boiling devices. Calculate the weighted average of the water boiling efficiency in the project boundary using the proportion of different stove types used and the stove efficiencies.
Source of data	- Baseline survey - Literature Review
Value(s) applied	- Three-stone fire without either a grate or a chimney: 10%. - Three-stone fire for charcoal: 16% - Improved cookstoves: Efficiency 37.9[15]%.
Choice of data or Measurement methods and procedures	-
Purpose of data	Baseline emissions calculation
Additional comment	-

Parameter ID	SDWS 12
Data/parameter	C_b
Unit	Percentage
Description	Proportion of project end-users who in the baseline were already using safe water, either from an improved water source, or from a water treatment method other than boiling
Source of data	Any of the following sources shall be used: - Baseline survey

15 This is the rated thermal efficiency of improved cookstove distributed by Community Carbon or conservative consideration of any cross-effects.

	- Credible published literature for project region - Studies by academia, NGOs or multilateral institutions - Official government publications or statistics Source applied must not be more than 3 years old.
Value(s) applied	0
Choice of data or Measurement methods and procedures	-
Purpose of data	Calculation of baseline emissions
Additional comment	-

Parameter ID	SDWS 8
Data/parameter	x_f
Unit	Percentage of fuel f use in target population
Description	The proportion of each different cooking fuel f used in the project boundary by end-users: % among the target population if single fuel is used for water boiling. For example, the target population either use wood or charcoal - 60% end users use wood and 40% charcoal. - Weighted average on energy basis, if multifuel situation exists within premise. If the project covers different types of end-users premises (e.g. households, schools), then the fuels used in the geographical area of the project by the same types of end-users are to be determined for each end-user premises type. Undertake assessment at the start of each crediting period.
Source of data	- Baseline survey Studies by academia, NGOs or multilateral institutions
Value(s) applied	Three stove fire – 27% Traditional Charcoal – 65% Improved Charcoal Cookstove – 8%%
Choice of data or Measurement methods and procedures	-

Purpose of data	-
Additional comment	-

Parameter ID	SDWS 21
Data/parameter	$f_{NRB,f,y}$
Unit	Percentage
Description	Fractional non-renewability status of woody biomass fuel during year y, in case the baseline fuel is biomass or charcoal
Source of data	Assessment based on CDM Methodological tool 30: Calculation of the fraction of non-renewable biomass, Version 03.0
Value(s) applied	78 %
Choice of data or Measurement methods and procedures	Assessment based on CDM Methodological tool 30: Calculation of the fraction of non-renewable biomass, Version 03.0
Purpose of data	CO_2 Emission calculation in project scenario
Additional comment	The f_{NRB} value will remain fixed during the crediting period

Parameter ID	SDWS 14
Data/parameter	NCV_f
Unit	TJ/fuel units, i.e. mass or volume units
Description	Net calorific value of fossil fuel f
Source of data	IPCC defaults
Value(s) applied	0
Choice of data or Measurement methods and procedures	-

Purpose of data	Calculation of project emissions
Additional comment	There is no use of fossil fuel in the project scenario. In case a new model is introduced then IPCC default values of that corresponding fuel will be used.

Parameter ID	SDWS 15
Data/parameter	EF_f
Unit	tCO2/TJ
Description	Emission factor of fossil fuel f
Source of data	IPCC defaults
Value(s) applied	0
Choice of data or Measurement methods and procedures	-
Purpose of data	Calculation of project emissions
Additional comment	There is no use of fossil fuel in the project scenario. In case a new model is introduced then IPCC default values of that corresponding fuel will be used.

Data/parameter	EF_{ec}
Unit	tCO2/kWh
Description	Emission factor associated with the electricity use
Source of data	Default values
Value(s) applied	0.01 tCO2/kWh if annual consumption is more than 250 kWh/year/household or institution

Choice of data or Measurement methods and procedures	-
Purpose of data	Calculation of project emissions
Additional comment	There is no use of electricity in the project device anticipated to be implemented under the VPA. In case a new model is introduced then default values will be used.

Parameter ID	SDWS 17
Data/parameter	TDL_{ec}
Unit	%
Description	Transmission and distribution losses associated with the electricity use
Source of data	Default values
Value(s) applied	20%
Choice of data or Measurement methods and procedures	-
Purpose of data	Calculation of project emissions
Additional comment	There is no use of electricity in the project device anticipated to be implemented under the VPA. In case a new model is introduced then default values will be used.

SDG 1: No Poverty

Data/parameter	$HHS_{baseline}$
Unit	Percentage (%)
Description	% users reporting money saving due to reduced fuel consumption in baseline
Source of data	Baseline survey

Value(s) applied	0
Choice of data or Measurement methods and procedures	--
Purpose of data	SDG 1 impact calculation

SDG 3: Good Health and Well Being

Data/parameter	$HHclean_{baseline}$
Unit	Percentage (%)
Description	% users confirming access to clean technology
Source of data	Baseline survey
Value(s) applied	0
Choice of data or Measurement methods and procedures	--
Purpose of data	SDG 3 impact calculation

SDG 5: Gender Equality

Data/parameter	$HHtime_{baseline}$
Unit	Percentage (%)
Description	% users reporting time savings due to reduced collected fuel consumption
Source of data	Baseline survey
Value(s) applied	0
Choice of data or Measurement methods and procedures	--

Purpose of data	SDG 5 impact calculation

SDG 6: Clean Water & sanitation

Data/parameter	$BENclean_{baseline}$
Unit	Number
Description	Beneficiaries with access to safe water due to project
Source of data	Baseline survey
Value(s) applied	0
Choice of data or Measurement methods and procedures	--
Purpose of data	SDG 6 impact calculation

SDG 8: Decent Work and Economic Growth

Data/parameter	$EG_{baseline}$
Unit	Number
Description	Employment and income generation
Source of data	Baseline survey
Value(s) applied	0
Choice of data or Measurement methods and procedures	--
Purpose of data	SDG 8 impact calculation

SDG 12: Responsible Consumption and Production

Data/parameter	$FC_{baseline}$
Unit	Percentage (%)
Description	% users confirming boiling using fuel in baseline scenario
Source of data	Baseline survey
Value(s) applied	100
Choice of data or Measurement methods and procedures	--
Purpose of data	SDG 12 impact calculation

SDG 15: Life on Land

Data/parameter	$FRC_{baseline}$
Unit	Percentage (%)
Description	% users reported fuelwood equivalent savings in baseline scenario
Source of data	Baseline survey
Value(s) applied	100
Choice of data or Measurement methods and procedures	--
Purpose of data	SDG 15 impact calculation

B.6.3. Ex ante estimation of SDG Impact

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Ex-ante calculations of emission reductions related to outcomes of SDG 13:

Emission Reductions

$ER_y = BE_y - PE_y - LE_y$

Where:

ER_y	Emission reductions in year y (t CO2e/yr)
PE_y	Project emissions in year y (tCO2)
BE_y	Baseline emissions in year y (t CO2e/yr)
LE_y	Leakage emissions in year y (t CO2e/yr)

Baseline Emission Reduction

The baseline emission factor shall be calculated as

$EF_b = SE_{w,b,y} * (\sum x_f * (EF_{b,f,CO2} * f_{NRB,f,y} + EF_{b,f,nonCO2})) \div 10^9$

$SE_{w,b,y} = 360.83/\eta_{wb} \quad = 360.83/12.09\% = 2984.53$ kJ/L

$EF_b = 2984.53 * [15\% * (165.22 * 78\% + 44.83) + 85\% (112 * 78\% + 9.46) \div 10^9$

$= 0.0003233$

The quantity of safe drinking water provided by the project Qy is determined as follows:

$QPW_{hh,p,y} = \min((q_i \times t_{p,y} \times DN_{p,y}), (QPW_p \times HN_{p,y}))$

$= \min((1800*7*2), (3*300)) = 900$

$Q_y = \sum_p N_{p,y} \times U_{p,y} \times QPW_{hh,p,y} \times DP_{p,y}$

$= 600*90\%*900*365$

$= 136.08$ M-litres

The baseline emission shall be calculated as

$BE_y = EF_b * (1 - C_b - X_{cleanboil,y}) \times Q_y \times M_{q,y}$

$= 0.0003233 * (1-0-0) * 136.08*1$

$= 44{,}006\ tCO_2e/yr$

Project Scenario Fuel Consumption Calculation

Project emissions shall be calculated as

$PE_y = PE_{ff,p,y} + PE_{ec,p,y}$

$PE_y = 0\ tCO_2e/yr$

In case the CME introduces safe water technology which uses electricity in future, project emissions shall be calculated according to the methodology.

Leakage

As explained under section B.6.1, for ex-ante ER estimation,

$LE_y = 0$

Finally,

$ER_y = BE_y - PE_y - LE_y$
$= 44{,}006 - 0 - 0$

$= 44{,}006\ tCO2e$

Average for 5 years = 26,477 tCO_2e

The detailed calculations of estimated ex-ante emission reductions in tCO_2e are provided in a separate excel calculation sheet.

SDG targeted	SDG impact indicator	Calculation	Ex-ante estimation
1 No poverty	**Indicator:** % users reporting money saving due to reduction in purchased fuel consumption in project	$HHS_{project}$-$HHS_{baseline}$ = 100% -0%	100%
3 Good health and Well Being	**Indicator:** % users HH confirming access to Clean Technology	$HHclean_{project}$-$HHclean_{baseline}$ = 100%-0%	100%
5 Equality	**Indicator**: % users reporting time saving due to reduction in collected fuel consumption / cooking time in project	$HHtime_{project}$-$HHtime_{baseline}$ =95%-0%	95%
6 Clean Water & Sanitation	**Indicator:** Users reporting access to WPS	$BENClean_{baseline}$ – $BENclean_{project}$ = 1,086,300- 0	1,083,00
8 Decent Work and economic growth	**Indicator:** Number of male / female numbers of employment created by project	$EG_{project}$ -$EG_{baseline}$ =40-0	40
12 Responsible consumption and production	**Indicator:** % users reporting fuel savings in the project	$FC_{baseline}$- $FC_{project}$ =100%-0%	100%

15 Life on Land	**Indicator:** % users reported fuelwood eq saving in the project scenario	$(FCHH_{project}-FRC_{baseline}$ =100%-0%	100%

B.6.4. Summary of ex ante estimates of each SDG outcome

SDG 13

Year	Baseline estimate	Project estimate	Net benefit
Year 1	367	0	367
Year 2	14,669	0	14,669
Year 2	29,337	0	29,337
Year 4	44,006	0	44,006
Year 5	44,006	0	44,006
Total	**132,385**	**0**	**132,385**
Total number of crediting years			
Annual average over the crediting period	26,477	0	26,477

SDG 1

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0%	100%	100%
Year 2	0%	100%	100%
Year 2	0%	100%	100%
Year 4	0%	100%	100%
Year 5	0%	100%	100%
Total	**0%**	**100%**	**100%**
Total number of crediting years	**5**		
Annual average over the crediting period	**0%**	**100%**	**100%**

SDG 3

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0%	100%	100%
Year 2	0%	100%	100%
Year 2	0%	100%	100%
Year 4	0%	100%	100%
Year 5	0%	100%	100%
Total	**0%**	**100%**	**100%**
Total number of crediting years	**5**		
Annual average over the crediting period	**0%**	**100%**	**100%**

SDG 5

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0%	95%	95%
Year 2	0%	95%	95%
Year 2	0%	95%	95%
Year 4	0%	95%	95%
Year 5	0%	95%	95%
Total	**0%**	**95%**	**95%**
Total number of crediting years	**5**		
Annual average over the crediting period	**0%**	**95%**	**95%**

SDG 6

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0	1,500	1,500
Year 2	0	60,000	60,000

Year 2	0	1,20,000	1,20,000
Year 4	0	1,80,000	1,80,000
Year 5	0	1,80,000	1,80,000
Total	**0**	1,80,000	1,80,000
Total number of crediting years	**5**		
Annual average over the crediting period	**0**	**1,80,300**	**1,80,300**

SDG 8

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0	40	40
Year 2	0	40	40
Year 2	0	40	40
Year 4	0	40	40
Year 5	0	40	40
Total	**0**	40	40
Total number of crediting years	**5**		
Annual average over the crediting period	**0**	**40**	**40**

Total number of jobs generated over 5 years = 70

SDG 12

Year	Baseline estimate	Project estimate	Net benefit
Year 1	100%	0%	100%
Year 2	100%	0%	100%
Year 2	100%	0%	100%
Year 4	100%	0%	100%
Year 5	100%	0%	100%
Total	**100%**	**0%**	**100%**

Total number of crediting years	**5**		
Annual average over the crediting period	**100%**	**0%**	**100%**

SDG 15

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0%	100%	100%
Year 2	0%	100%	100%
Year 2	0%	100%	100%
Year 4	0%	100%	100%
Year 5	0%	100%	100%
Total	**0%**	**100%**	**100%**
Total number of crediting years	**5**		
Annual average over the crediting period	**0%**	**100%**	**100%**

B.7. Monitoring plan

B.7.1. Data and parameters to be monitored

SDG 13

Parameter ID	SDWS 24
Data/parameter	QPWp
Unit	Liters/person/day
Description	Volume of drinking water per person per day for premises type p
Source of data	For ex-ante calculation default values are used, but CME may choose to monitor the parameter or use default value during verification. Option 1: Apply the default value per person. Option 2: Water Consumption Field Tests. - In all cases, the value is capped at 5.5 L/person/day

Value(s) applied	For ex-ante ER: Residential (full-day premise) - 4 Institutional (assumed half-day premises) – 3
Monitoring methods and procedures	In case of Option-2 is chosen during verification, water consumption field tests will be carried out for 3-days and the values would be free of outliers and conservativeness would be demonstrated.
Monitoring frequency	Every two years
QA/QC Procedures	Water consumption field tests would be compared with qualitative response of the users on the usage frequency/quantity and household size.
Purpose of data	Calculation of baseline emissions
Additional comment	In case of Option-2, minimum sample size for HWT is 30 and for IWT CDM Guideline for sampling and surveys will be used to determine sample size.

Parameter ID	SDWS 13
Data/parameter	q_i
Unit	Litres per hour
Description	Capacity of the household or institutional water treatment technology
Source of data	- Commercial guarantee by the seller
Value(s) applied	1800 litres/hour (for ex-ante estimation)
Monitoring methods and procedures	-
Monitoring frequency	Continuous
QA/QC Procedures	-
Purpose of data	Calculation of baseline emissions
Additional comment	This depends on water filtration device model and is fixed for each model introduced. In case different model is introduced the value may differ.

Parameter ID	SDWS 22
Data/parameter	$X_{cleanboil,y}$
Unit	Percentage
Description	Proportion of project end-users that boil safe (treated, or from safe supply) water after installation of project technology in year y.
Source of data	Project survey.
Value(s) applied	0 (for ex-ante ER)
Monitoring methods and procedures	Representative survey will be carried out to determine this parameter
Monitoring Frequency	Annually
QA/QC Procedure	-
Purpose of data	Estimation of CO_2e emission reductions
Additional comment	-

Parameter ID	SDWS 18
Data/parameter	$M_{q,y}$
Unit	Fraction
Description	Ongoing water quality indicated as the fraction of the samples that pass microbial quality standard requirements specified in relevant microbial quality standard for drinking water of the host country. In case a national standard is not available, the water quality shall comply with WHO Guideline values for verification of microbial quality i.e., all water directly intended for drinking must not have detectable E.Coli in any 100 ml sample i.e., less than 1 Colony Forming Unit (CFU) of E.Coli /100 ml

Source of data	Testing of water all appliances or at a representative sample of end-users at the following point where the water reaches the end-user premises: HWT and IWT: testing of the water that exits the treatment technology.
Value(s) applied	1
Measurement methods and procedures	Field testing kits using Most Probable Number (MPN) method will be used.
Monitoring Frequency	Annually
QA/QC procedures:	Testing kits shall be tested for its accuracy and robustness prior to application for project level monitoring
Purpose of data	Estimation CO2 emissions
Additional comment	If the proportion of samples not meeting Safe Drinking Water Quality Standards exceeds a permissible limit of <1 Colony Forming Unit(CFU)/100 ml[16] no emission reductions will be claimed for the corresponding monitoring period. A minimum sample size of 30 will be selected.

Parameter ID	SDWS 28
Data/parameter	$N_{p,y}$
Unit	Number
Description	Accumulated number of premises type p with at least one individual project technology in year y
Source of data	Sales or distribution records
Value(s) applied	600
Measurement methods and procedures	-

[16] https://www.cdc.gov/infectioncontrol/guidelines/environmental/appendix/water.html

Monitoring Frequency	Annually
QA/QC procedures	. Sales or distribution records to include: i. Date of sale/distribution ii. Geographic area of sale iii. Model/type of project technology sold iv. Quantity of project technologies sold Name and telephone number, and address (if available) or other traceable indicator of premises identity and location for all end users.
Purpose of data	Estimation of CO_2e emission reductions
Additional comment	Yearly distribution scheduled indicative is mentioned in the Emission reduction calculation. The actual implementation could be different based on demand.

Parameter ID	SDWS 29
Data/parameter	$U_{p,y}$
Unit	percentage
Description	Usage rate of the project technology by premises type p during year y
Source of data	Survey the premises with a project technology to determine the usage rate of the project technology during the year. Option 1: In-person survey of project premises Households that show at least once-in-two-days use may be counted as users. The resulting fraction is multiplied by 100% to get$U_{p,y}$.
Value(s) applied	90% (Assumed for ex-ante calculation)
Measurement methods and procedures	In-person representative survey
Monitoring Frequency	Annually

QA/QC Procedures	Where a WCFT is undertaken to determine QPWp, this may be used to cross check the usage percentage.
Purpose of data	Estimation of CO_2e emission reductions
Additional comment	The usage survey provides a single usage parameter that is representative for project technologies in the total sales record. Applies to all HWT and IWT technologies and projects The minimum sample size for HWT - for individual technology age group shall be minimum 30 household.]

Parameter ID	SDWS 31
Data/parameter	$DP_{p,y}$
Unit	Days
Description	Average days the project technology is present for end-users in the premises p in year y
Source of data	Sales or distribution records. Based on the sales or distribution records of "Date of sale/distribution" and ex-ante parameter "Expected technical life of project technology," determine for each project device how many days of the 365 days of the year it was in the premises and within its technical life. Calculate the average for all the project technology by premises type p to obtain this parameter.
Value(s) applied	Residential – 365 Institution- 280 (for ex-ante)
Measurement methods and procedures	-
Monitoring Frequency	Annually
QA/QC Procedures	For schools and other institutions, as applicable, the days must also be limited by the number of school days in the period, taking into account weekends and holidays.
Purpose of data	Estimation of CO_2e emission reductions
Additional comment	-

Parameter ID	SDWS 30
Data/parameter	$t_{p,y}$
Unit	Hours per day
Description	Usage time of the project technology by premises type p in year y
Source of data	Determined via Project survey Option 1. Observational sample-based survey of project household practices.
Value(s) applied	7 (for ex-ante estimation)
Measurement methods and procedures	Representative survey
Monitoring frequency	Annually
QA/QC Procedures	
Purpose of data	Estimation of CO_2e emission reductions
Additional comment	-

Parameter ID	SDWS 32
Data/parameter	$DN_{p,y}$
Unit	Number
Description	Average number of individual project technologies in each project premises type p in year y
Source of data	Sales or distribution records.
Value(s) applied	2
Measurement methods and procedures	Based on the sales or distribution records of “Quantity of project technologies sold”, the average number of project devices per premises. If the project covers different types of end-users (e.g. households, institutions), the average number will be determined per premises type p.
Monitoring Frequency	Annually
QA/QC Procedures	NA

Purpose of data	Estimation of CO_2e emission reductions
Additional comment	-

Parameter ID	SDWS 25
Data/parameter	$HN_{p,y}$
Unit	Number
Description	Number of individuals per premises type p in the project boundary in year y
Source of data	Any of the following sources shall be used: - Project survey - Official government publications or statistics - Credible published literature for project region, or - Studies by academia, NGOs or multilateral institutions Source applied must not be more than 3 years old.
Value(s) applied	300 (Average school size for Malawi for ex-ante estimation)
Measurement methods and procedures	-
Monitoring Frequency	Annually
QA/QC Procedures	-
Purpose of data	Estimation of CO_2e emission reductions
Additional comment	-

Parameter ID	SDWS 33
Data/parameter	$P_{p,f,y}$
Unit	mass or volume units (e.g. kg, Litres, standard m3)
Description	Quantity of fossil fuel f that is consumed in the project during year y
Source of data	Any of the following methods will be used: - Direct measurement with meter, scales - Estimation with e.g. tank capacity table - Taken from fuel invoice or purchase receipt, or

	- In the case of direct fuel use by water treatment systems, may be estimated from the manufacturer's specification of the equipment and operating hours or volumes (e.g. fuel consumption per hour times utilization hours or fuel consumption per litre times the litres of water treated).
Value(s) applied	0 (Gravity filters don't use any electricity or fossil fuel in project scenario)
Measurement methods and procedures	Any of the following methods will be used: - Direct measurement with meter, scales - Estimation with e.g. tank capacity table - Taken from fuel invoice or purchase receipt, or - In the case of direct fuel use by water treatment systems, may be estimated from the manufacturer's specification of the equipment and operating hours or volumes (e.g. fuel consumption per hour times utilization hours or fuel consumption per litre times the litres of water treated)
Monitoring frequency	Continuously
QA/QC Procedures	If measured, devices should be calibrated.
Purpose of data	Estimation of CO_2e emission reductions
Additional comment	-

Parameter ID	SDWS 34
Data/parameter	$EC_{p,y}$
Unit	KWh
Description	Quantity of electricity that is used by the project during year y
Source of data	Any of the following methods will be used: - Direct measurement with electric meter - Sample based using electricity loggers, or - In the case of direct electricity use by water treatment systems, may be estimated from the manufacturer's specification of the equipment and operating hours or volumes (e.g. electricity consumption per hour times utilization hours or electricity consumption per litre times the litres of water treated).
Value(s) applied	0 (Gravity filters don't use any electricity or fossil fuel in project scenario)

Measurement methods and procedures	Any of the following methods will be used: - Direct measurement with electric meter - Sample based using electricity loggers, or - In the case of direct electricity use by water treatment systems, may be estimated from the manufacturer's specification of the equipment and operating hours or volumes (e.g. electricity consumption per hour times utilization hours or electricity consumption per litre times the litres of water treated).
Monitoring frequency	Continuously
QA/QC Procedures	If measured, devices should be calibrated.
Purpose of data	Estimation of CO_2e emission reductions
Additional comment	-

Parameter ID	SDWS 20
Data / Parameter	Water hygiene education campaigns
Unit	-
Description	Hygiene campaigns carried out among project safe water end-users.
Source of data	Report of annual hygiene campaigns results
Value(s) applied	Report of the annual hygiene campaign
Measurement methods and procedures	-
Monitoring frequency	Annually
QA/QC procedures	The fraction of the households where Safe water and Hygiene practices are found to fulfill "safely managed" or "basic" requirements is expected to increase over time as a result of the hygiene campaigns.
Purpose of data	Estimation of CO_2e emission reductions and monitoring of SDG 12
Additional comment	-

Parameter ID	SDWS 35

Data/parameter	$LE_{,y}$
Unit	tCO_2/year
Description	Leakage emissions during year y
Source of data	Survey
Value(s) applied	0 (for ex-ante ER estimation)
Measurement methods and procedures	Representative survey will be carried out to determine this parameter.
Monitoring frequency	Every two years
QA/QC Procedures	Compliance with the general requirements for sampling and general requirements for data and information sources.
Purpose of data	Estimation of CO_2e emission reductions
Additional comment	-

SDG 1: End poverty;

Data / Parameter	$HHS_{project}$
Unit	Percentage (%)
Description	% users reporting money saving due to reduction in purchased fuel consumption in project
Source of data	Representative survey/CME Database
Value(s) applied	100
Measurement methods and procedures	Survey
Monitoring frequency	Annual
QA/QC procedures	-
Purpose of data	Monitoring of SDG 1
Additional comment	-

SDG 3: Good Health and Well-being

Data / Parameter	$HHclean_{project}$
Unit	Percentage (%)
Description	% users using clean technology to access safe water in project scenario
Source of data	Representative survey/CME Database
Value(s) applied	100
Measurement methods and procedures	The number of water purification devices distributed is recorded as part of the CME database which will be used to estimate this parameter
Monitoring frequency	Continuous
QA/QC procedures	-
Purpose of data	Monitoring of SDG 3
Additional comment	-

SDG 5: Gender Equality

Data / Parameter	$HHtime_{project}$
Unit	Percentage (%)
Description	% users reporting time saving associated with cooking and fuel collection
Source of data	Monitoring Survey
Value(s) applied	95%
Measurement methods and procedures	This parameter will be monitored as part of the monitoring survey. Users will be asked as part of monitoring survey if they have been able to save time due to use of clean energy products
Monitoring frequency	Once every 2 years (biennially)
QA/QC procedures	-
Purpose of data	Monitoring of SDG 5
Additional comment	The parameter will be measured qualitatively.

SDG 6: Clean Water and Sanitation

Data / Parameter	$BENclean_{project}$
Unit	Number
Description	SDG 6.1.1 Contributions (i) Level of Service: Safely managed service (ii) Project contributions Number of beneficiaries
Source of data	CME Database Water Quality Tests
Value(s) applied	108300
Measurement methods and procedures	The total number of water purification systems distributed to average family member from project survey is used to estimate this parameter.
Monitoring frequency	Annual
QA/QC procedures	-
Purpose of data	Monitoring of SDG 6
Additional comment	-

SDG 8: Decent Work and Economic Growth

Data / Parameter	$EG_{project}$
Unit	Number
Description	Generation of employment in project scenario
Source of data	Employee List
Value(s) applied	40
Measurement methods and procedures	CME will maintain a record of jobs created as a result of implementation of this programme in form employee list/ payroll system/ contracts/ pay slips
Monitoring frequency	Continuous
QA/QC procedures	-
Purpose of data	Monitoring of SDG 8

Additional comment	-

SDG 12: Responsible Consumption and Production

Data / Parameter	$FC_{project}$
Unit	Percentage (%)
Description	% users confirming boiling using fuel in project scenario
Source of data	Monitoring survey
Value(s) applied	0
Measurement methods and procedures	Survey
Monitoring frequency	Annually
QA/QC procedures	-
Purpose of data	Monitoring of SDG 12
Additional comment	-

SDG 15: Life on Land

Data / Parameter	$FRC_{baseline}$
Unit	Percentage (%)
Description	% users reported fuelwood (eq.) savings in project scenario
Source of data	Monitoring survey
Value(s) applied	0
Measurement methods and procedures	Survey
Monitoring frequency	Annually
QA/QC procedures	-
Purpose of data	Monitoring of SDG 15
Additional comment	-

B.7.2. Sampling plan

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Cross-VPA sampling to be explored (again based on scale of project). Random sampling will be preferred in case of multiple technologies and vintage. It will be ensured more than 10 VPAs are not grouped while carrying out cross VPA sampling.

The sampling approach is to be used for following surveys as summaries below:

A statistically valid sample can be used to determine parameter values, as per the relevant requirements for sampling in the "Methodology for Sampling and surveys for CDM project activities and programme of activities." Minimum 90% confidence interval and a 10% margin of error requirement shall be achieved for the sampled parameters. In any case, for proportion parameter values, a minimum sample size of 30, or the whole group size if this is lower than 30, must always be applied. Further, cross-VPA sampling is not accepted across groups larger than 10 VPAs. In case of cross VPA sampling, 95/10 confidence/precision will be applied.

When a baseline and project survey is used the following sample size guidelines should be applied, unless otherwise stated for specific parameters:

Water quality testing

- The sample for water quality testing will be made following the 90/10 precision rule indicated by the applied methodology.

Usage Survey:

Group size	Minimum sample size
<300	30 or population size, whichever is smaller
300 to 1000	10% of group size
>1000	100

Sampling Plan for monitoring parameters

The objective of the sampling plan for this VPA is to determine:

i. $U_{p,y}$: Usage rate of the project technology by premises type p during year y. This would be based on usage survey, to be conducted upon the representative sample of operational technology.

ii. $M_{q,y}$: Water quality. Testing of water at a representative sample of endusers where the water reaches the end-user premises.

iii. $X_{cleanboil,y}$: The actual value will be determined based survey to be conducted upon representative sample.

iv. QPW_p: Volume of drinking water per person per day for premises type p. This be based on Option1 or Option2 opted by PP.

v. $t_{p,y}$: Usage time of the project technology by premises type p in year y. This be based on Option1 or Option2 or Option 3 opted by PP.

vi. $DN_{p,y}$: Average number of individual project technologies in each project premises type p in year y. The actual value will be determined based survey to be conducted upon representative sample.

B.7.3. Other elements of monitoring plan

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Monitored Systems

1. **Total Distribution Record:** The total distribution record documents the information listed below for the technologies implemented. A carbon waiver including a warranty card will be distributed with each technology (HWT and/or IWT) distributed. The CME makes every effort to retrieve this information (paper form or electronically (i.e., SMS) but cannot guarantee the collection of information for waivers and warranties with every technology due to challenges such as high rates of illiteracy and logistical challenges. The total distribution record will be kept electronically and with supporting evidence from paper records and/or SMS tracking records and will be provided to the GS-VVB at verification. The Total Distribution Record contains:
    a) VPA-ID (VPA to which appliance belongs)
    b) Unique identification of WPS using WPS serial number
    c) Partner organization name, address and telephone (as available)
    d) Date of distribution and model/type of project technology distributed
    e) Quantity of project technology distributed as evidenced by invoices

Frequency: Ongoing

2. **Project Database:** Each VPA will have a specific Project Database that records each WPS crediting in that VPA. Every WPS listed in the Total Distribution Record will be transferred into the Project Database of this VPA as needed to expand the number of WPS deployed, until the maximum threshold for this VPA is reached. In addition to the information provided in the Total Distribution Record, the VPA-specific Project Database will record user details (enough for end-user identification and follow-up) for all, or a subset of all, appliances deployed. End-user details recorded are:
    a) Name
    b) Telephone, or address (as available)
    c) Baseline technology: whether boiling or no boiling
    d) Type of WPS (WPS model) and fuel the WPS is replacing: Example – traditional or improved baseline stoves, or wood or charcoal fuel.

HWT or IWT with end-user details recorded here will be used to determine other information needed using as many samples as commensurate with representative sampling to calculate the emission reductions of the project.

HWT or IWT with end-user details recorded here will be used to determine other information needed using as many samples as commensurate with representative sampling to calculate the emission reductions of the project

Frequency: Ongoing

3. **Continued use of displaced baseline technology**

The replaced stoves used for boiling are encouraged to be disposed of and not used within the boundary or within the region. Monitoring surveys conducted on households (HH) using HWT/IWT will also investigate the extent to which baseline stoves are still in use.

4. **Organizational structure of monitoring and inclusions:**

Person	Role
CME database administrator	The database administrator is responsible for updating and maintaining all electronic databases. Required competencies

	include experience with data management systems (e.g., Excel, STATA, or SPSS), minimum 2 years working experience in a similar field, and at minimum a bachelor's degree from an institution of higher education.
Monitoring team	The monitoring team will be assigned by the CME to conduct the user interviews and appliance tests during the periodic sampling and reports the results to the database administrator. The skills and experience required for the data collection activities include: • Experience conducting surveys/tests • Experience conducting door-to-door surveys of biomass consumption • Local language skills (especially important for input to questionnaire design and interviewing of end users) • English language skills • Cultural awareness • Numerical proficiency • Data entry skills

5. Quality Assurance / Quality control

As the PoA is intended to include multiple regions within a country with a high level of cultural diversity as well as different end user groups, there is no "one size fits all" approach for dealing with these issues. However, in order to avoid many of these problems the CME will undertake the following strategies, tailoring the specific approach to the local circumstances:

a) Ensuring end user awareness. At the time of distribution, the HWT/IWT customer is made aware that they are required to participate in monitoring activities. This will be via a written statement (in English and local language where appropriate) on the carbon waiver form, or via alternative means such as training distribution personnel to explain the importance of monitoring to each customer.

b) Questionnaire design. The design of the questionnaire will ensure that the questions are non-intrusive and easy to understand for both the interviewee and interviewer.
c) Drawing on local knowledge. The local contractors to be hired by the CME in the country will play an important role in tailoring the approach to suit local circumstances.
d) Quality of contractors. Any third parties hired by the CME to carry out sampling will be required to demonstrate a high level of cultural awareness, local language skills and appropriate experience with data entry and data management. The CME will ensure that contractors are adequately trained for the tasks they are contracted for (e.g., carrying out of field tests in line with a methodology supported by an appropriate international body/standards). Training will also be provided on how to deal with non-responses, refusals and other problems should these occur.

6. Hygiene Campaign

The hygiene campaign will be designed as per the guidelines laid out in the methodology. Water hygiene education campaigns will include the following activities:

a) Educating customers on proper handling and storage of raw and processed water.
b) Emphasizing the importance of hygiene to prevent infections or water borne diseases. The training campaigns would be organised during the product demonstration and repeated on twice in year so that customers remember.
c) Designing survey questionnaire for water filters to include all the necessary questions related to the project in line with JMP Core questionnaire by WHO/UNICEF. On annual basis, customers will be assessed to check the effectiveness of the education campaign. Customers will be selected randomly from the total population.

The project will report the activities conducted each year in the annual monitoring report. Any major changes in the health status of the water users as a result of contaminated water (e.g. an outbreak of water related disease) will be reported and, if relevant, a strategy put in place to address it through the hygiene campaign.

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1. Start date of VPA

>>

15/06/2022

C.1.2. Expected operational lifetime of VPA

>>

15 years (3 x 5 years)

C.2. Crediting period of project

C.2.1. Start date of crediting period

>>

14/09/2022

C.2.2. Total length of crediting period

>>

5 years. The crediting period may be renewed twice in line with the Community Services Activity Requirements.

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in Appendix 1, no ongoing monitoring is applicable to this VPA.

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The local stakeholder consultation meeting will be carried out following a gender sensitive approach. The project will incorporate measures to ensure that there is no discrimination based on gender. HWT/IWT will be distributed to all willing customers within

	the project boundary. The project will have a positive impact on women considering that they will spend less time on boiling water for treatment or walking for miles to collect water and fuel.	
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The Malawi National Gender Policy 2015 provides guidelines for mainstreaming gender in various sectors of the economy with the overall goal of reducing gender inequalities and enhancing participation of women, men, girls and boys in socio economic and political development. The purpose of the policy is to strengthen gender mainstreaming and women empowerment at all levels in order to facilitate attainment of gender equality and equity in Malawi. The project will contribute towards the goal of policy by providing women opportunity to save time used in either collecting fuelwood for boiling water to make it fit for drinking or travelling long distance to collect drinking water.	
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	Not required. Safe Water Supply projects not following Gender responsive approach do not require to contract an expert as per Gender Equality Requirements & Guidelines. The project will have a positive impact on women included under the programme by providing them with water purification systems requiring minimal manual work.	
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	Not required. Safe Water Supply projects not following Gender responsive approach do not require to contract an expert as per Gender Equality Requirements & Guidelines. The stakeholder consultation will include interactions with potential beneficiaries including women and their feedback shall be recorded appropriately.	

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

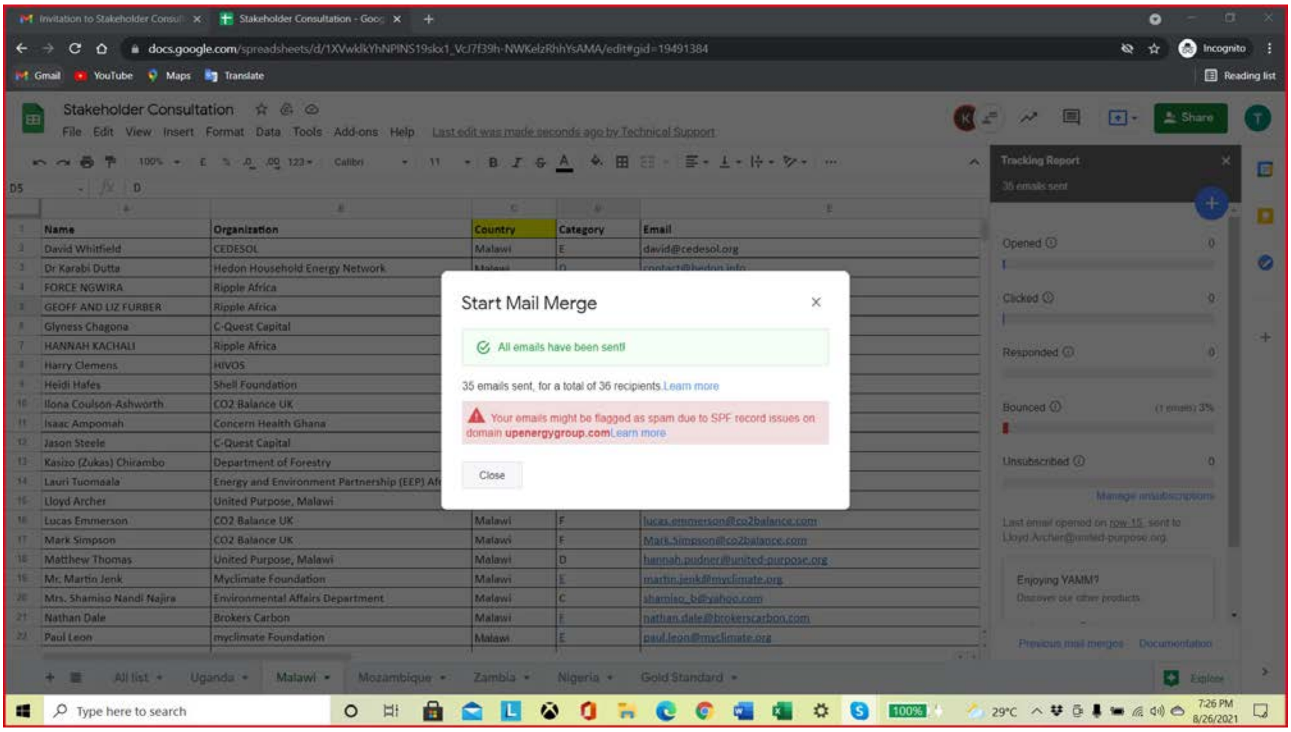
The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1. Summary of stakeholder mitigation measures

>>

The physical local stakeholder consultation meeting was conducted on 25th September 2021 (which is more than/at least 30-days' notice given to the stakeholders). Invitations were sent to stakeholders along with feedback form and non-technical summary via email on 26th August 2021. Also, newspaper advertisement was published on 23rd August 2021 and 16th September 2021. The screenshots of the advertisement are shown below. Apart from e-mail and newspaper advertisement, stakeholders were invited via informal channels such as phone, whatsapp. The feedback forms were distributed during the physical meeting as well as via email along with invitation.

1. Email Invitation sent on 26th August 2021



--------- Forwarded message ---------
From: **Carbon Technical Team** <technical@upenergygroup.com>
Date: Thu, Aug 26, 2021 at 7:22 PM
Subject: Invitation to Stakeholder Consultation Meeting for UpEnergy (Malawi)– Social and Climate Impact Programme
To: Technical <technical@upenergygroup.com>

Dear David Whitfield ,

Greetings from The UpEnergy Group.

UpEnergy Group is the Co-ordinating/Managing entity of GS ID 10963, UpEnergy – Social and Climate Impact Programme

The PoA (and its VPAs 1 to 14) is registered under Global Standard for the Global Goals (GS4GG) to certify the contribution of PoA towards achievement of various SDGs.

The purpose of the Gold Standard Programme of Activities (PoA) is to provide end-users with energy-efficient cooking technologies and Safe water systems (WPS) with the aim to reduce greenhouse gas (GHG) emissions from the burning of non-renewable woody biomass and/or charcoal for cooking and water treatment to make it safe for consumption. Thus, the PoA supports the intended goals of reducing fuel consumption, improving health, reducing deforestation and reducing GHG emissions in Uganda

As a valued stakeholder of the PoA, you are requested to participate in the design of an initiative intended to create, register and trade carbon credits in order to fund the expanded dissemination of improved cooking stoves and water purification systems.

The stakeholder consultation is an opportunity at the early stage to present the projects to the relevant parties involved and include their feedback into the design of the projects.

We are currently beginning the initial stakeholder consultation process. You are requested to attend the physical meeting which will be held on **25/09/2021 at Riverside Hotel & Conference centre, Paul Kagame Road, near Lingadzi inn, Area 32, Lilongwe, Malawi**. We welcome your feedback and suggestions and cordially request you to fill the form to share your views on this PoA.

A brief non-technical description of the projects is available upon request by email, and a working draft of the Project Design Document will be also publicly available for revision soon. The written feedback can be sent via email to technical@upenergygroup.com up to 7 days after the consultation day.

Thank you for your consideration of this request. We look forward to receiving your comments and/or having the pleasure of meeting you in person on 25/09/2021.

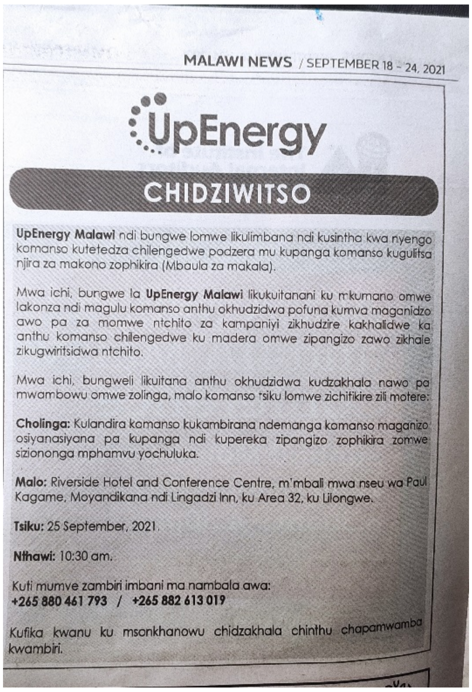
Thanking you,

With Regards

UpEnergy Group

Lilongwe, Malawi

2. Newspaper advertisement on 18th September 2021


MALAWI NEWS / SEPTEMBER 18 – 24, 2021

UpEnergy

CHIDZIWITSO

UpEnergy Malawi ndi bungwe lomwe likulimbana ndi kusintha kwa nyengo komanso kutetedza chilengedwe podzera mu kupanga komanso kugulitsa njira za makono zophikira (Mbaula za makala).

Mwa ichi, bungwe la **UpEnergy Malawi** likukuitanani ku mkumano omwe lakonza ndi magulu komanso anthu okhudzidwa pofuna kumva maganidzo awo pa za momwe ntchito za kampaniyi zikhudzire kakhalidwe ka anthu komanso chilengedwe ku madera omwe zipangizo zawo zikhale zikugwiritsidwa ntchito.

Mwa ichi, bungweli likuitana anthu okhudzidwa kudzakhala nawo pa mwambowu omwe zolinga, malo komanso tsiku lomwe zichitikire zili motere:

Cholinga: Kulandira komanso kukambirana ndemanga komanso maganizo osiyanasiyana pa kupanga ndi kupereka zipangizo zophikira zomwe sizionongа mphamvu yochuluka.

Malo: Riverside Hotel and Conference Centre, m'mbali mwa nseu wa Paul Kagame, Moyandikana ndi Lingadzi Inn, ku Area 32, ku Lilongwe.

Tsiku: 25 September, 2021.

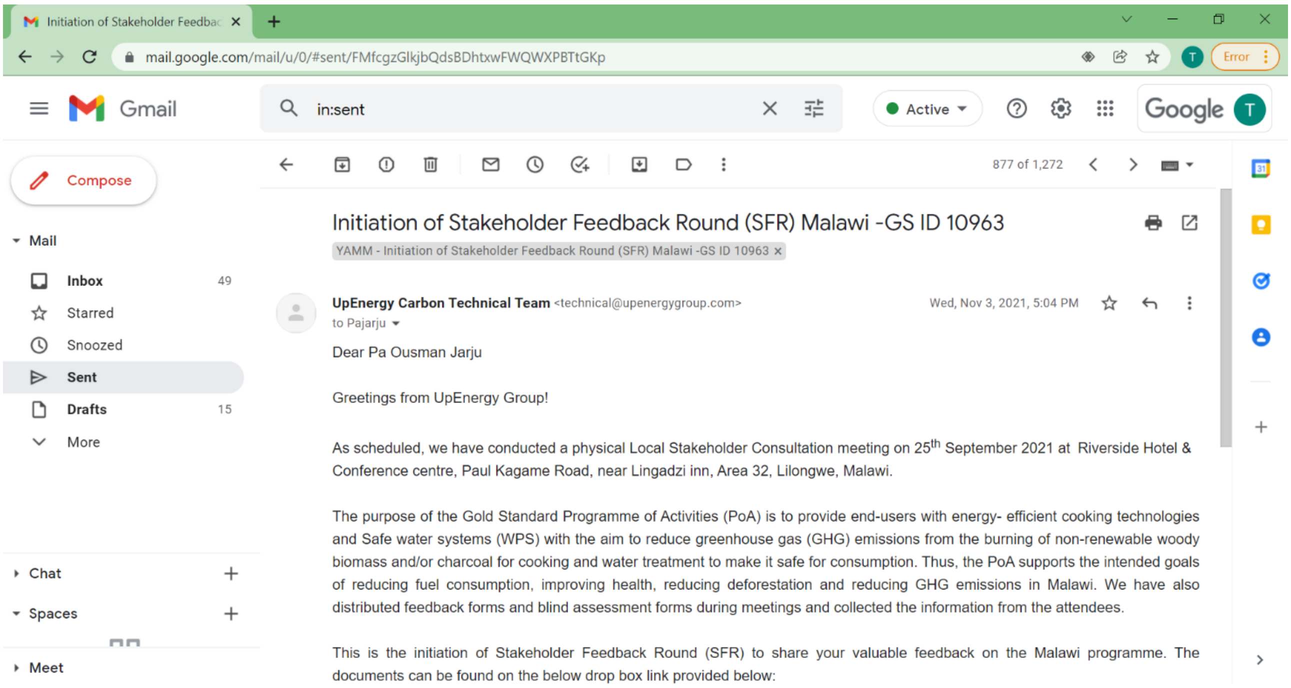
Nthawi: 10:30 am.

Kuti mumve zambiri imbani ma nambala awa:
+265 880 461 793 / +265 882 613 019

Kufika kwanu ku msonkhanowu chidzakhala chinthu chapamwamba kwambiri.

The SFR was initiated on 22nd September 2021. The reminder email for SFR closure was sent on 15th December 2021. The SFR closing mail was sent on 3rd January 2022. The screenshots of the emails are below:

1. SFR Initiation Mail on 22nd September 2021



Initiation of Stakeholder Feedback Round (SFR) Malawi -GS ID 10963

YAMM - Initiation of Stakeholder Feedback Round (SFR) Malawi -GS ID 10963

UpEnergy Carbon Technical Team <technical@upenergygroup.com> Wed, Nov 3, 2021, 5:04 PM

to Pajarju

Dear Pa Ousman Jarju

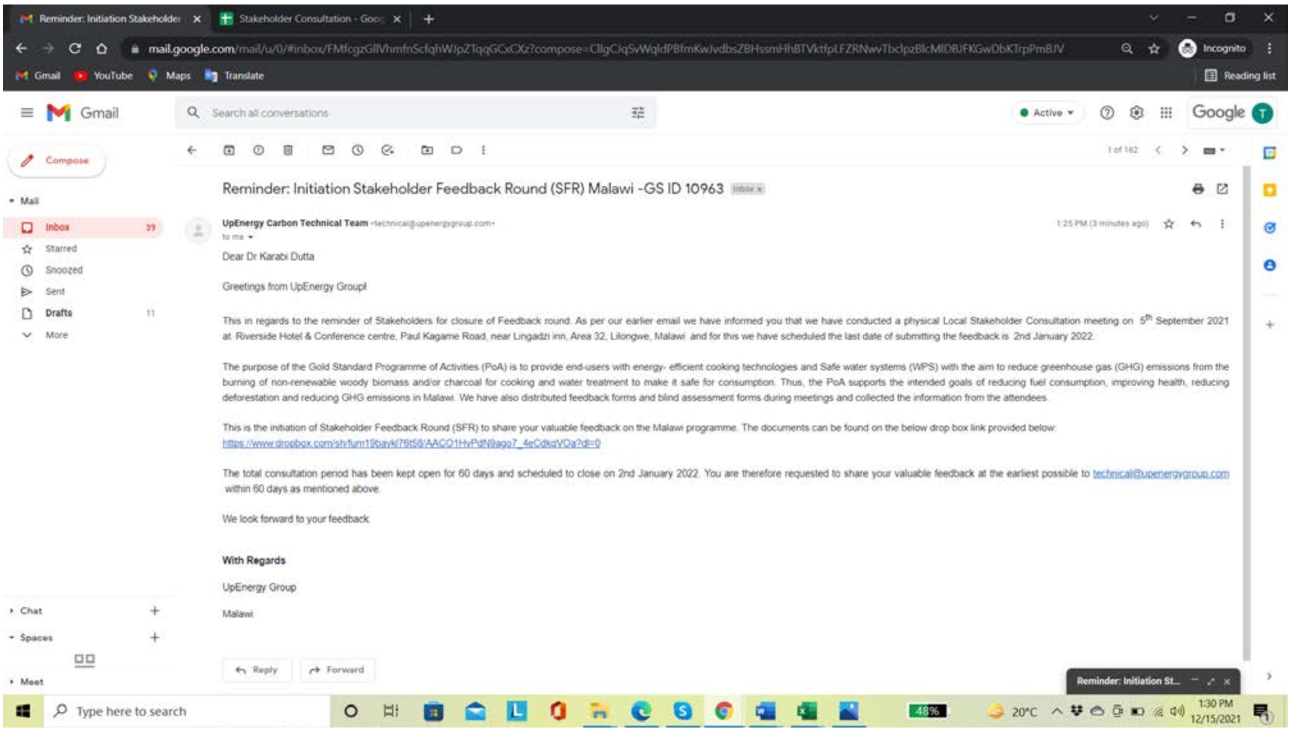
Greetings from UpEnergy Group!

As scheduled, we have conducted a physical Local Stakeholder Consultation meeting on 25th September 2021 at Riverside Hotel & Conference centre, Paul Kagame Road, near Lingadzi inn, Area 32, Lilongwe, Malawi.

The purpose of the Gold Standard Programme of Activities (PoA) is to provide end-users with energy- efficient cooking technologies and Safe water systems (WPS) with the aim to reduce greenhouse gas (GHG) emissions from the burning of non-renewable woody biomass and/or charcoal for cooking and water treatment to make it safe for consumption. Thus, the PoA supports the intended goals of reducing fuel consumption, improving health, reducing deforestation and reducing GHG emissions in Malawi. We have also distributed feedback forms and blind assessment forms during meetings and collected the information from the attendees.

This is the initiation of Stakeholder Feedback Round (SFR) to share your valuable feedback on the Malawi programme. The documents can be found on the below drop box link provided below:

2. Reminder mail sent on 15th December 2021



Reminder: Initiation Stakeholder Feedback Round (SFR) Malawi -GS ID 10963

UpEnergy Carbon Technical Team <technical@upenergygroup.com> 1:25 PM (3 minutes ago)

to me

Dear Dr Karabi Dutta

Greetings from UpEnergy Group!

This in regards to the reminder of Stakeholders for closure of Feedback round. As per our earlier email we have informed you that we have conducted a physical Local Stakeholder Consultation meeting on 5th September 2021 at Riverside Hotel & Conference centre, Paul Kagame Road, near Lingadzi inn, Area 32, Lilongwe, Malawi and for this we have scheduled the last date of submitting the feedback is 2nd January 2022.

The purpose of the Gold Standard Programme of Activities (PoA) is to provide end-users with energy- efficient cooking technologies and Safe water systems (WPS) with the aim to reduce greenhouse gas (GHG) emissions from the burning of non-renewable woody biomass and/or charcoal for cooking and water treatment to make it safe for consumption. Thus, the PoA supports the intended goals of reducing fuel consumption, improving health, reducing deforestation and reducing GHG emissions in Malawi. We have also distributed feedback forms and blind assessment forms during meetings and collected the information from the attendees.

This is the initiation of Stakeholder Feedback Round (SFR) to share your valuable feedback on the Malawi programme. The documents can be found on the below drop box link provided below:
https://www.dropbox.com/sh/fum19bayki76t58/AACO1HvPdN9ago7_4eCdkqVOa?dl=0

The total consultation period has been kept open for 60 days and scheduled to close on 2nd January 2022. You are therefore requested to share your valuable feedback at the earliest possible to technical@upenergygroup.com within 60 days as mentioned above.

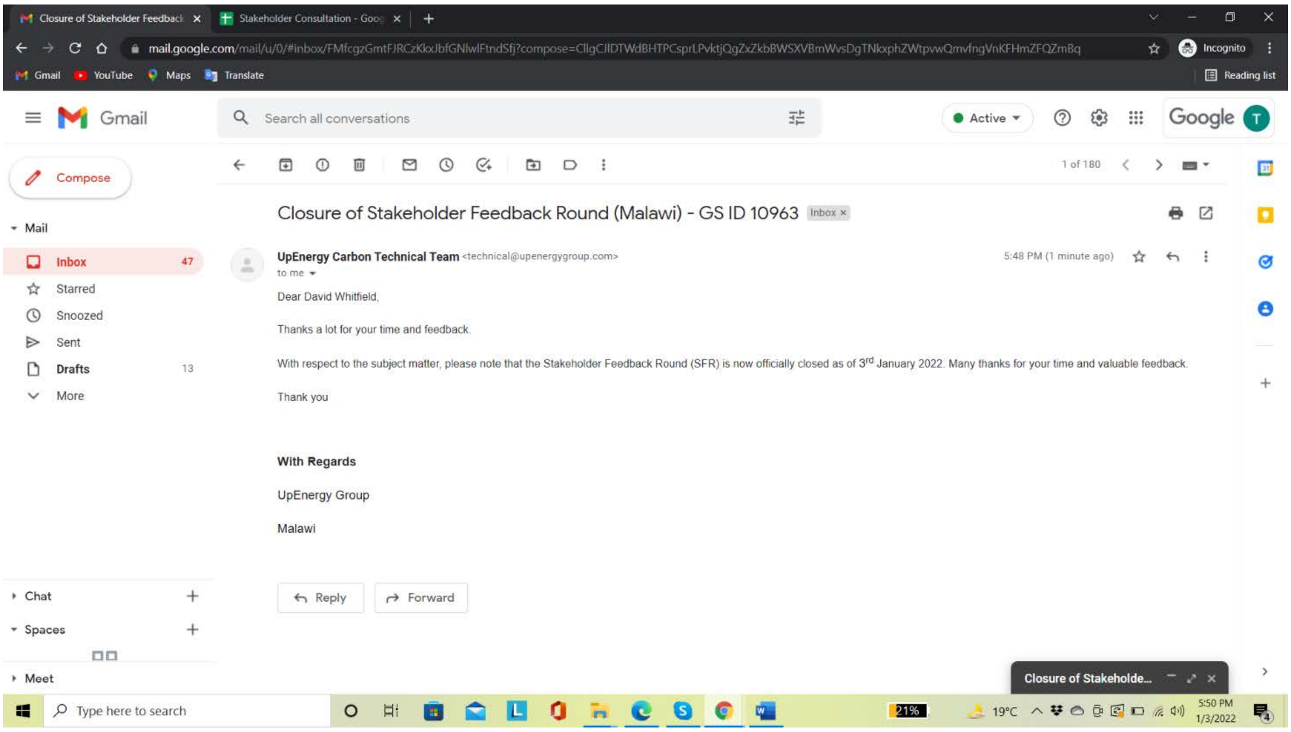
We look forward to your feedback.

With Regards

UpEnergy Group

Malawi

3. Closure email sent on 3rd January 2022



Three of our aspirational products such as the Domestic SmartHome Stove, Domestic and Institutional Safe Drinking Water System were displayed and a demonstration on how to use them was done. It was also explained to the stakeholders that similar technologies may also be implemented by Community Carbon based on beneficiary's feedback.

The consultation involved inviting comments/feedback from the following, but limited to, category of stakeholders:

- Local people impacted by the project and official representatives.
- Local policy makers and representatives of local authorities
- Local non-governmental organizations working on topics relevant to the PoA
- Host country DNA
- Research organizations/institutes working on the technology involved in the PoA
- Renewable energy/energy development nodal agencies
- GS NGO Supporters (present in that host country) + International GS NGO Supporter NGOs
- Other relevant local NGOs of the host country

The consultation process included invitation to wide range of invitees to include effective and equal participation of both men and women. But due to COVID, the attendees were less. Prior to consultation, Community Carbon had provided with the following documents in the language that allows local stakeholders to understand and engage with the project.

- Non-Technical summary with relevant information
- Summary of the economic, social and environmental impacts of the project
- Contact details of the Community Carbon group to get for getting technical and project related information of the project
- Email id of Community Carbon was provided to the stakeholders who are not able to join the consultation meeting

The stakeholder consultation report along with findings have been submitted separately. Summary of the few findings from LSC and SFR are listed below:

Organization	**Stakeholder Comments (Relevant)**	**If any changes in project design were made**
FEDOMA	Programme provides access to low income households with affordable products. The efficient stoves also leads to reduction in charcoal usage which in turn helps poor households save money.	NA
End User	Programme should provide good maintenance service to customers once the products are sold	NA
Lighthouse	Programme might create job opportunities for people	NA
End User	Programme has a follow up mechanism with end users	NA

	which ensure better adoption and impact from the programme	
KCH Hospital	Programme can bring in lot of health benefits to end users	

E.2. Final continuous input / grievance mechanism

Method	Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.
Continuous Input / Grievance Expression Process Book (mandatory)	In line with section 2.1 of the Annex W Expression book has been placed at office of UpEnergy Malawi Ltd in Malawi. Stakeholders are free to voice their concerns via the Grievance Expression Book. By maintaining feedback book at the local office, it is ensured that stakeholders that don't have access to electronic media for expressing concerns / grievances are also able to share their concerns / feedback. Additionally, the end users always have an option to revert to the salesperson (representative of distribution/retail partners etc.) in case of any feedback / complaints with the product post distribution.
GS Contact (mandatory)	help@goldstandard.org
Other	Email: anantha@communityco2.

SECTION F. Eligibility and inclusion criteria for VPAs inclusion

>>

Please Refer Section A.1.1

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into SECTION D above.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights 2. The Project shall not discriminate with regards to participation and inclusion	No	The project will be implemented in collaboration with local partners and CME will respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Right. The project will not discriminate with regards to participation and inclusion.	N/A
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse	No.	The project activity doesn't endorse any form of discrimination based on	NA

impacts on gender equality and/or the situation of women 2. Projects shall apply the principles of nondiscrimination, equal treatment, and equal pay for equal work 3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks 4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)		gender. Water Purification Systems (WPS) will be distributed to all willing customers within the project boundary. The project will have a positive impact on women considering that they will spend less time on boiling water for treatment or walking for miles to collect water and fuel.	
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	No	The project doesn't expose the community to increased health risks and is not adversely affecting the health of workers and the community. Use of WPS will contribute in improving the health of users as compared to baseline by reducing the chances of water-	N/A

		borne diseases through an efficient and zero GHG emission method. The workers participating in the project activity are not exposed to unsafe or unhealthy work environments as the sale/distribution of WPS or the monitoring activities of the project will not include any hazardous chemicals or other hazardous material.	
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	Since this is a WPS project, there is no risk of risk to cultural and historic heritage.	N/A
>>			
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples	No	Since this is a WPS project, there is no risk of forced eviction and displacement.	N/A

(temporary or permanent, full or partial)?			
>>			
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	Since this is a WPS project, there is no risk of forced eviction and displacement.	N/A
>>			
Principle 4.3 Land Tenure and Other Rights			
Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership?	No	Since this is a WPS project, there is no risk of uncertainty due to land rights/ownership.	N/A
>>			
Principle 4.4 - Indigenous people			
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?	**No**	The project does not affect indigenous population directly or indirectly	**Not required.**

Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	No	CME will ensure that the project doesn't involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects.	N/A
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions 2. Workers shall be able to establish and join labour organisations 3. Working agreements with all individual workers shall be documented and implemented and include:	No	1. The project is implemented on the ground by the UPE in collaboration with other project partners. The project employment will be in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions. 2. The workers employed by CME for the project are able to establish and join labour organizations. 3. The working agreements with the individual workers will be documented and	

a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labour is allowed (Exceptions for children working on their families' property requires an Expert Stakeholder opinion)		implemented and the minimum requirements stated will be respected as applicable. 4. The minimum age for possible staff hired is 18. 5. All the workers will be provided with appropriate equipment, training documentation and reporting of accidents and incidents, and emergency preparedness and response measures.	

5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures			
Principle 6.2 Negative Economic Consequences			
1. Does the project cause negative economic consequences during and after project implementation? >>	No	The project involves sale of WPS to willing customers within the project boundary. Carbon revenues are important for creating awareness among the end users and strengthening the local sales and distribution services.	N/A
Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario? >>	No	The project will reduce the GHG emissions which will be monitored and verified in line with the GS4GG.	N/A
Principle 7.2 Energy Supply			

Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?	No	The project does not use energy from a local grid or power supply. Use fuelwoond/ charcoal for boiling water in baseline using traditional stoves will be significantly reduced by introducing WPS.	N/A
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	The project is a WPS distribution programme and will not affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity.	N/A
>>			
Principle 8.2 Erosion and/or Water Body Instability			
Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion?	No	The project is a WPS distribution programme and will not directly or indirectly cause additional erosion and/or water body instability	N/A
>>			

		or disrupt the natural pattern of erosion.	
Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products? >>	No	The project is a WPS distribution programme and does not involve the use of land and soil for production of crops or other products.	N/A
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions? >>	No	The project is a WPS distribution programme and will not be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions.	N/A
Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms	No	The Project is not negatively impacted by the use of genetically modified organisms or GMOs.	N/A

that include GMOs in their processes and production)?			
>>			
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	No	The Project is a WPS distribution programme which are zero GHG emission products and does not result in the release of pollutants to the environment	N/A
>>			
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?	No	The Project does not involve the manufacture, trade, release, and/or use of hazardous chemicals and or materials.	N/A
>>			
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?	No	The project does not involve the application of pesticides and/or fertilisers.	N/A
>>			
Principle 9.7 Harvesting of Forests			

Will the Project involve the harvesting of forests?	No	The project does not involve the harvesting of forests.	N/A
>>			
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	No	The project does not modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives.	N/A
>>			
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	The project does not involve animal husbandry.	N/A
>>			
Principle 9.10 High Conservation Value Areas and Critical Habitats			
Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?	No	The project is a WPS distribution programme and physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified.	N/A
>>			
Principle 9.11 Endangered Species			

Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)? AND/OR Does the Project potentially impact other areas where endangered species may be present through transboundary affects?	No	The project boundary is geographical sites of WPS distributed and there are no endangered species identified as potentially being present within the Project boundary.	N/A
>>			

APPENDIX 2- CONTACT INFORMATION OF VPA IMPLEMENTER

Organization name	Community Carbon
Registration number with relevant authority	188181
Street/P.O. Box	
Building	
City	PORT LOUIS
State/Region	
Postcode	
Country	MAURITIUS
Telephone	
E-mail	anantha@communityco2.org
Website	
Contact person	
Title	DIRECTOR
Salutation	
Last name	RAGNAR BURNING
Middle name	
First name	KRISTIAN
Department	
Mobile	
Direct tel.	
Personal e-mail	

APPENDIX 3- LUF ADDITIONAL INFORMATION

Risk of change to the Project Area during Project Certification Period:	NA
Risk of change to the Project activities during Project Certification Period:	NA
Land-use history and current status of Project Area:	NA
Socio-Economic history:	NA
Forest management applied (past and future)	NA
Forest characteristics (including main tree species planted)	NA
Main social impacts (risks and benefits)	NA
Main environmental impacts (risks and benefits)	NA
Financial structure	NA
Infrastructure (roads/houses etc):	NA
Water bodies:	NA
Sites with special significance for indigenous people and local communities - resulting from the Stakeholder Consultation:	NA
Where indigenous people and local communities are situated:	NA
Where indigenous people and local communities have legal rights, customary rights or sites with special cultural, ecological, economic, religious or spiritual significance:	NA

APPENDIX 4-SUMMARY OF APPROVED DESIGN CHANGES

Please refer to Design Changes Requirements for more information on procedures governing Design Changes